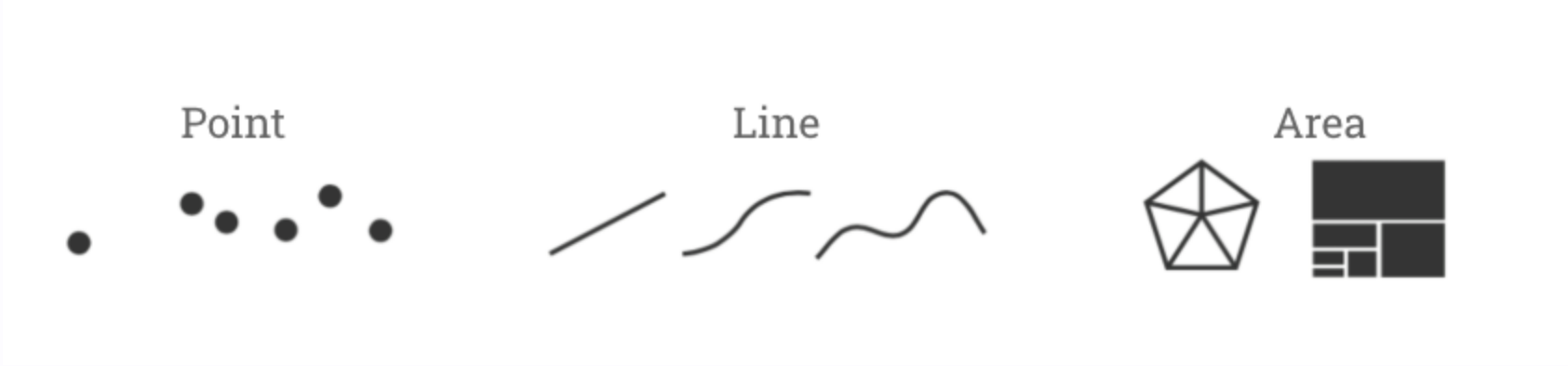
Language of Data Visualization

Marks and Channels



Marks ・　マーク

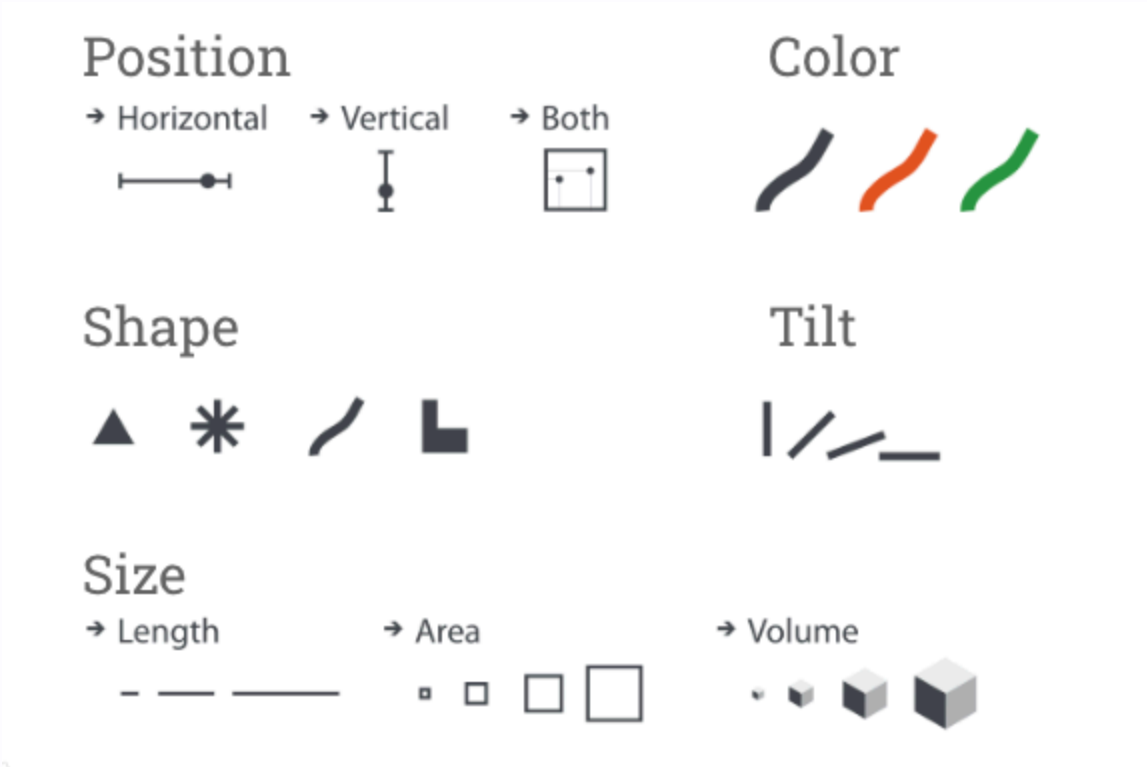
The shapes that represent data.
データを表す形

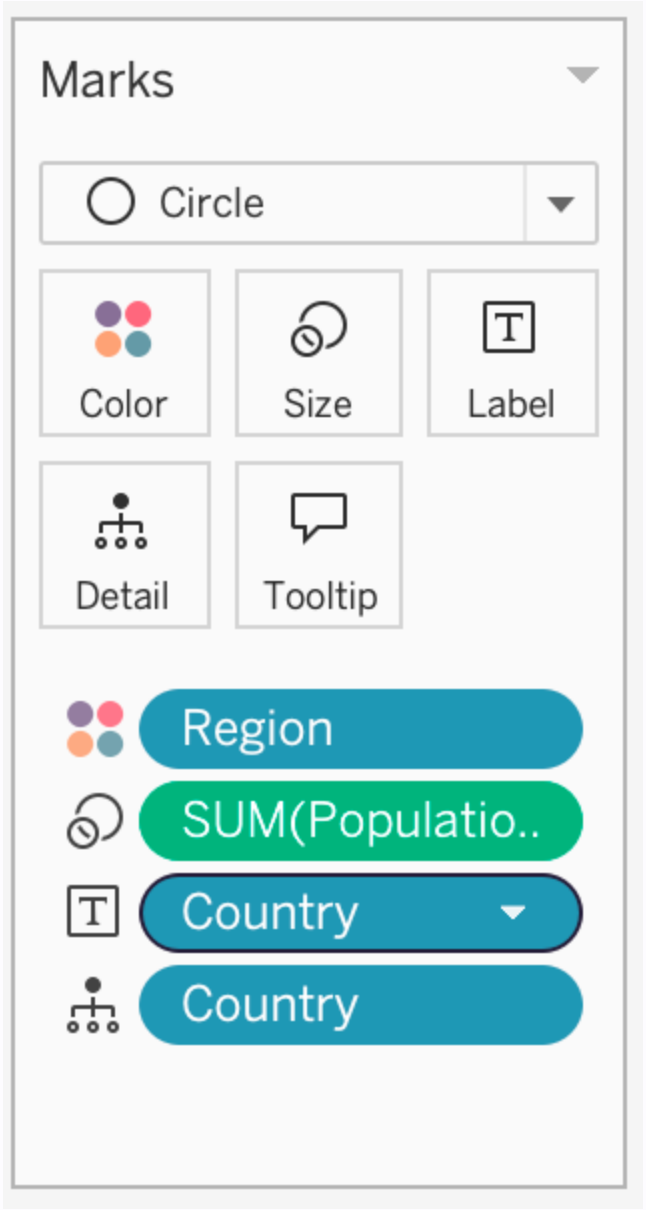


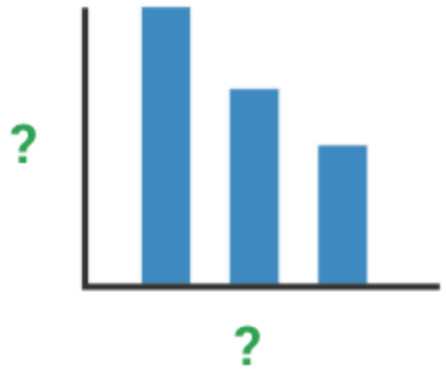
Graphics from Munzner. Visualization Analysis and Design (2015)

Channels ・ チャネル

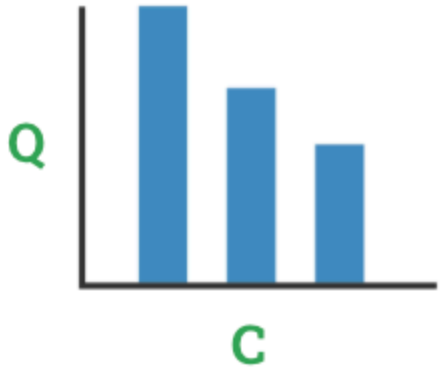
Ways to vary the appearance of marks
マークの見た目を変える方法



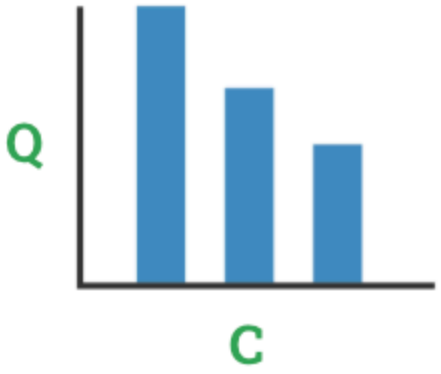

Marks
Circle
Color
Size
Label
Detail
Tooltip
Region
SUM(Populatio..
Country
Country



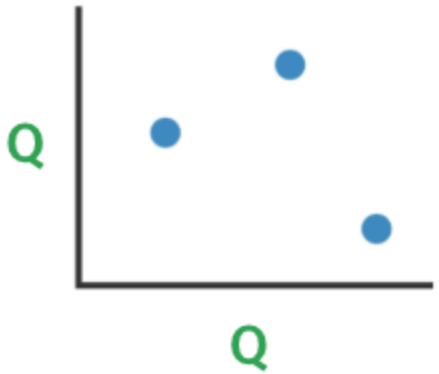
Mark: ?



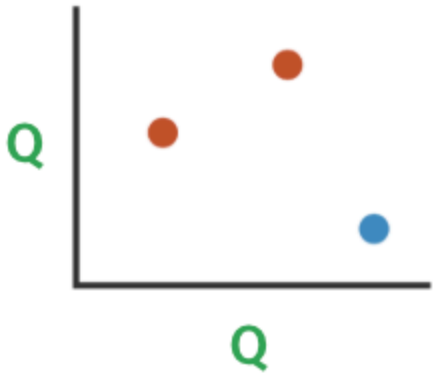
Mark: **line**



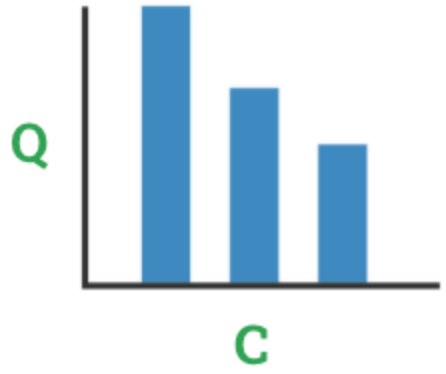
Mark: **line**



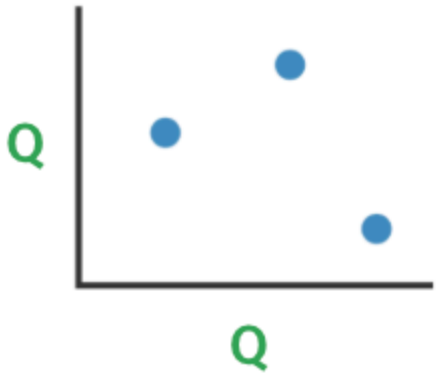
Mark: **point**



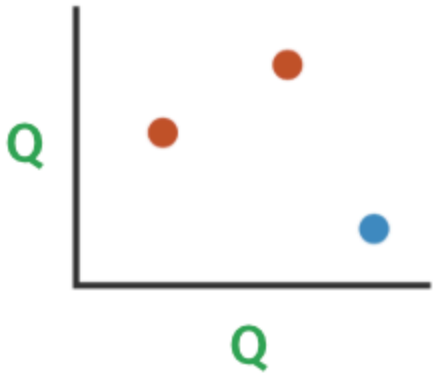
Mark: **point**
Channels:
Color: **C**



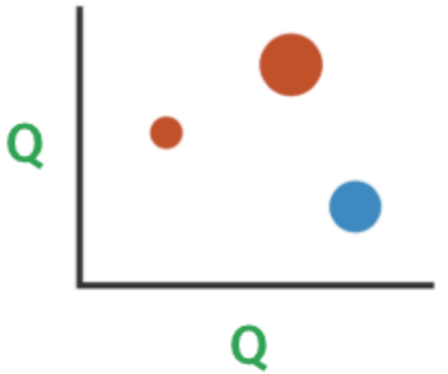
Mark: **line**



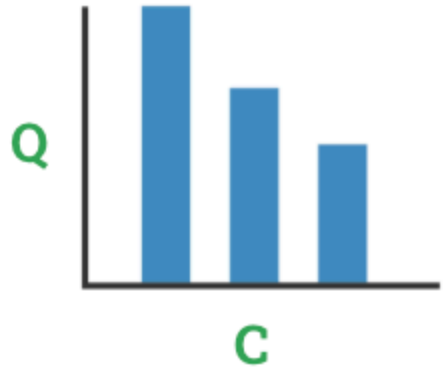
Mark: **point**



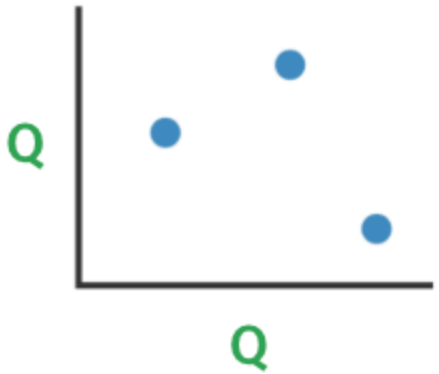
Mark: **point**
Channels:
Color: **C**



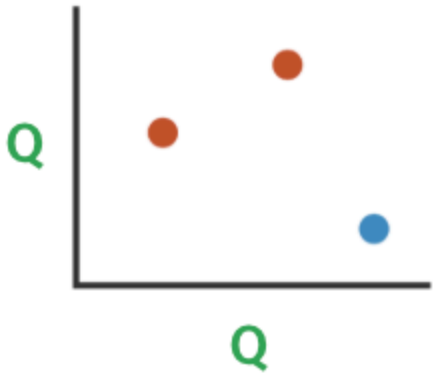
Mark: **point**
Channels: **?**



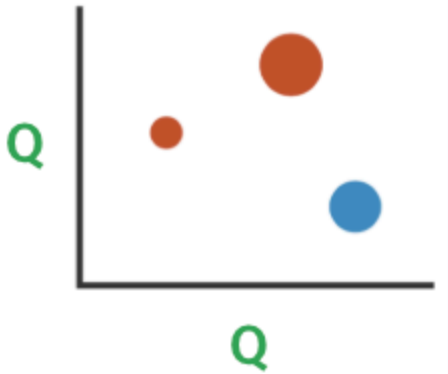
Mark: **line**



Mark: **point**



Mark: **point**
Channels:
Color: **C**



Mark: **point**
Channels:
Color: **C**
Size: **Q**

Position

→ Horizontal → Vertical → Both



Color



Shape



Tilt



Size

→ Length → Area → Volume



Position

→ Horizontal → Vertical → Both



Color



Categorical



Shape



Tilt



×

Ordinal



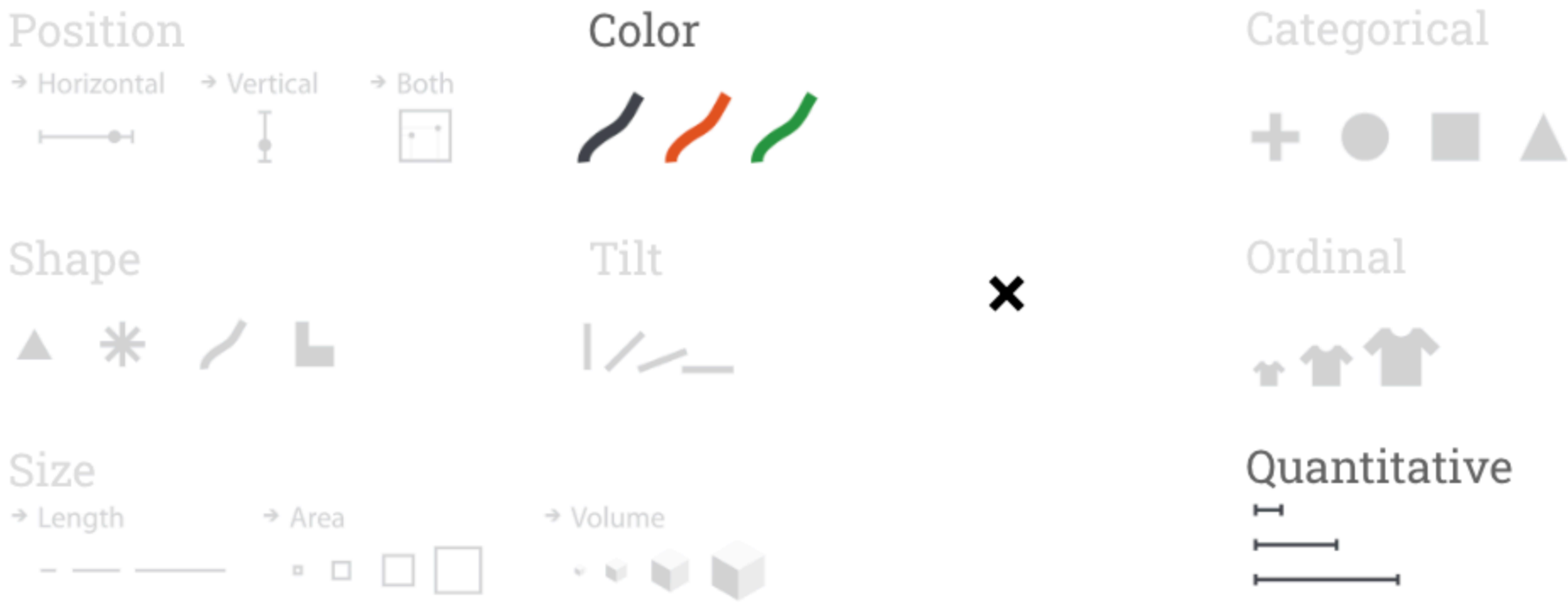
Size

→ Length → Area → Volume



Quantitative




Position
Horizontal
Vertical
Both
Color
Categorical
Shape
Tilt
Ordinal
Size
Length
Area
Volume
Quantitative

Filters

Marks

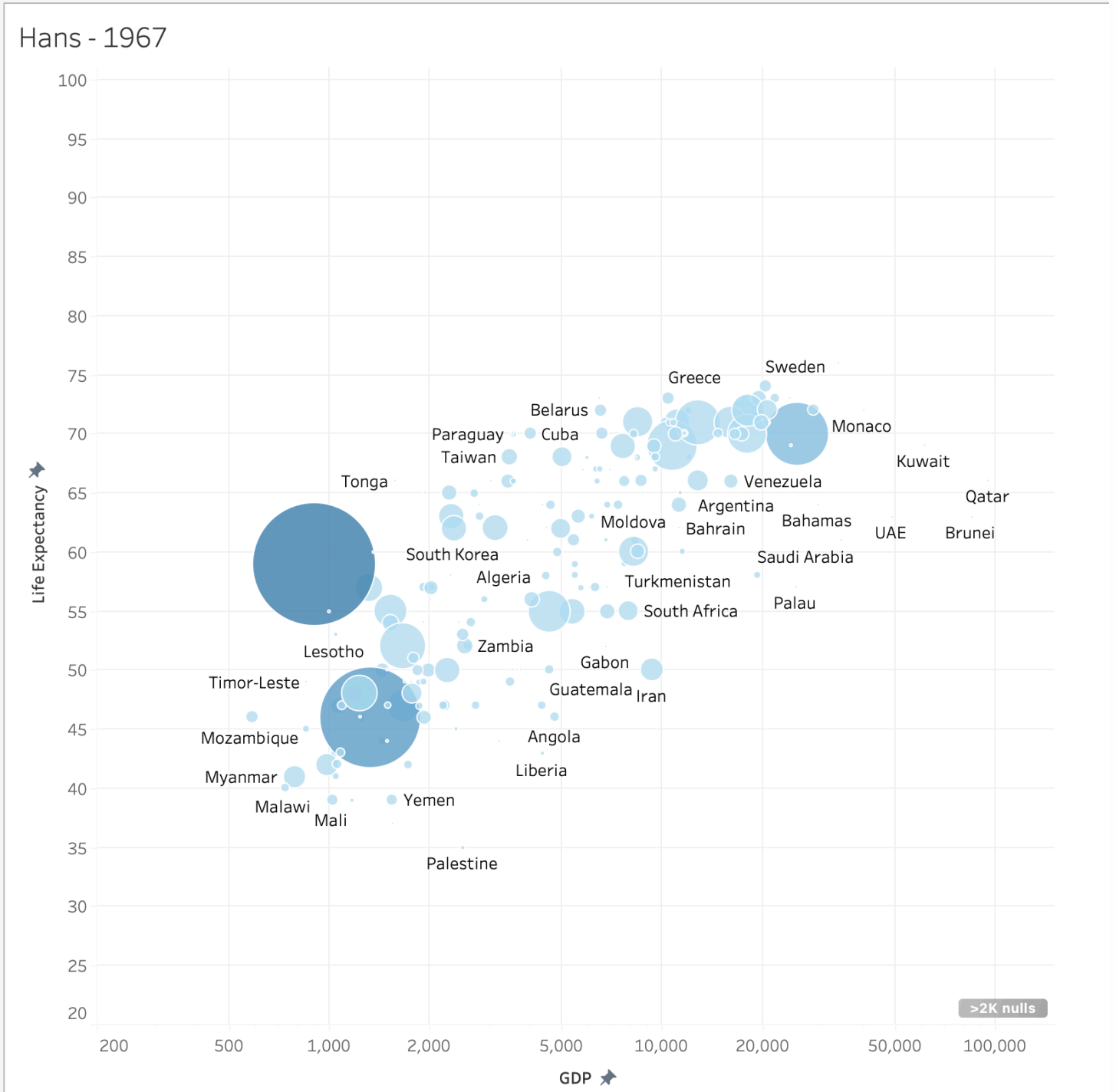
Circle

Color | Size | Label | Detail | Tooltip

SUM(Populatio..
SUM(Populatio..
Country
Country

SUM(Population)

651 999M



Position

→ Horizontal → Vertical → Both

Color



Categorical



Shape

Tilt

×

Ordinal

Size

→ Length → Area → Volume

Quantitative

Filters

Marks

Circle

Color | Size | Label

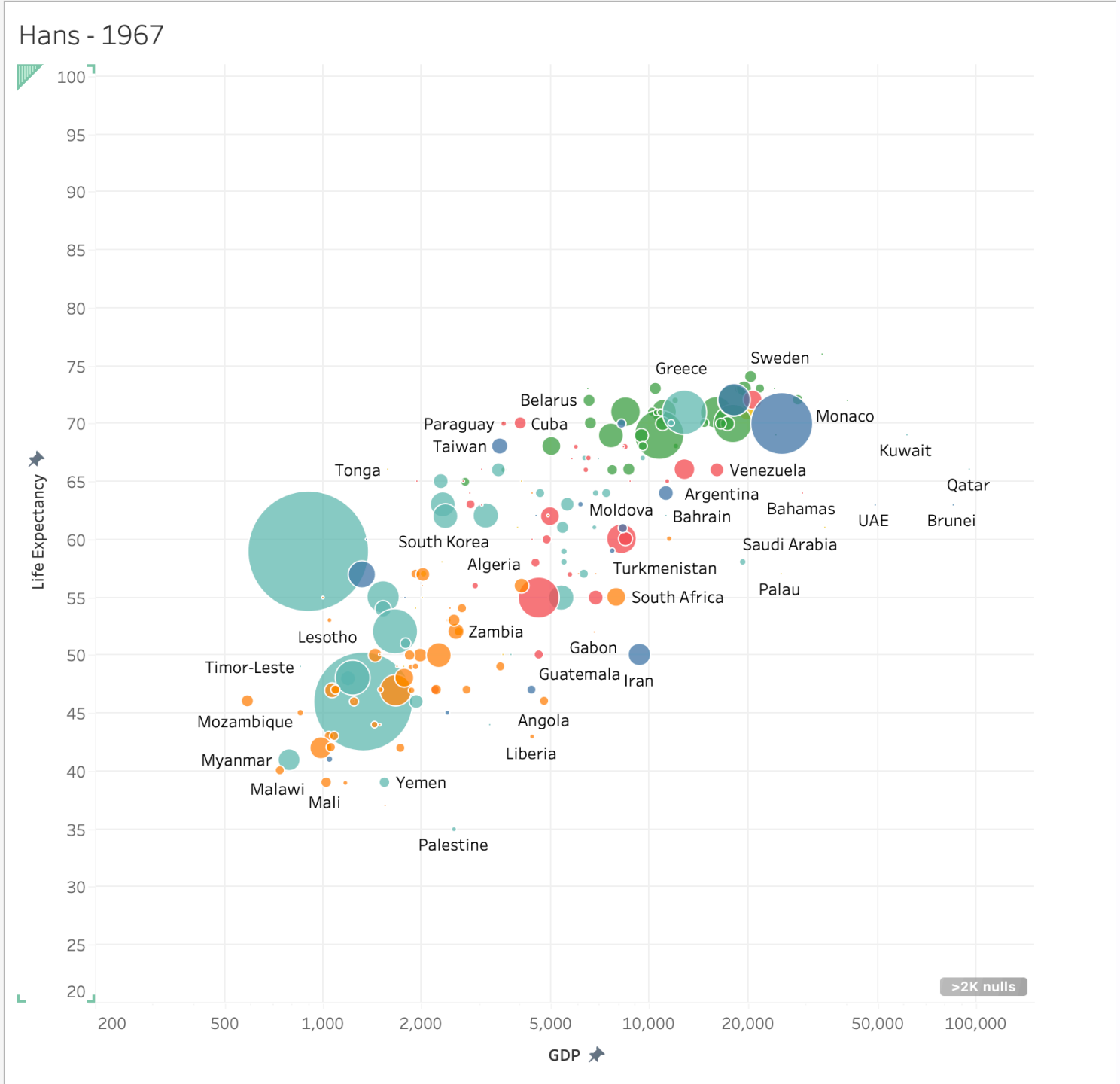
Detail | Tooltip

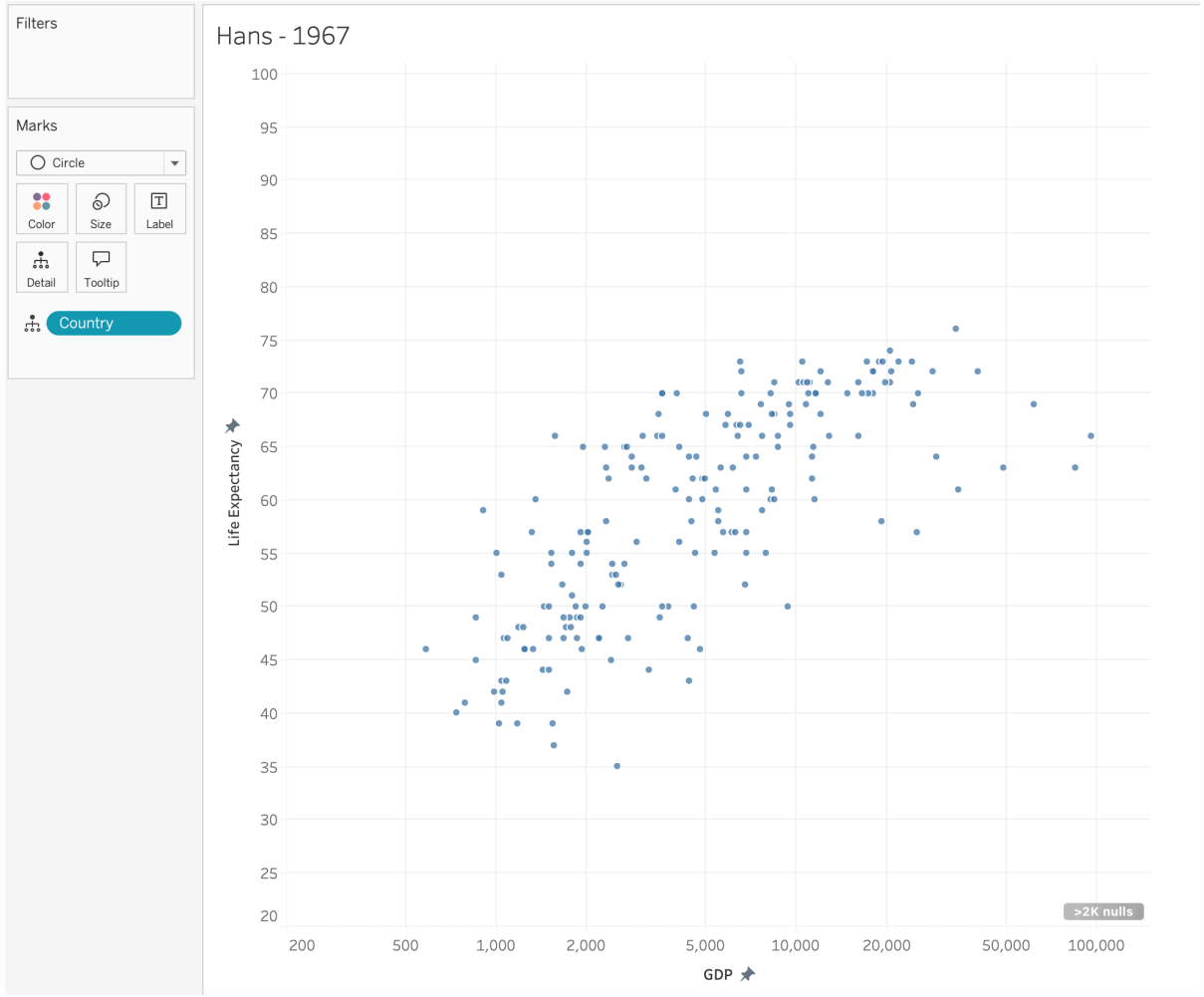
Region

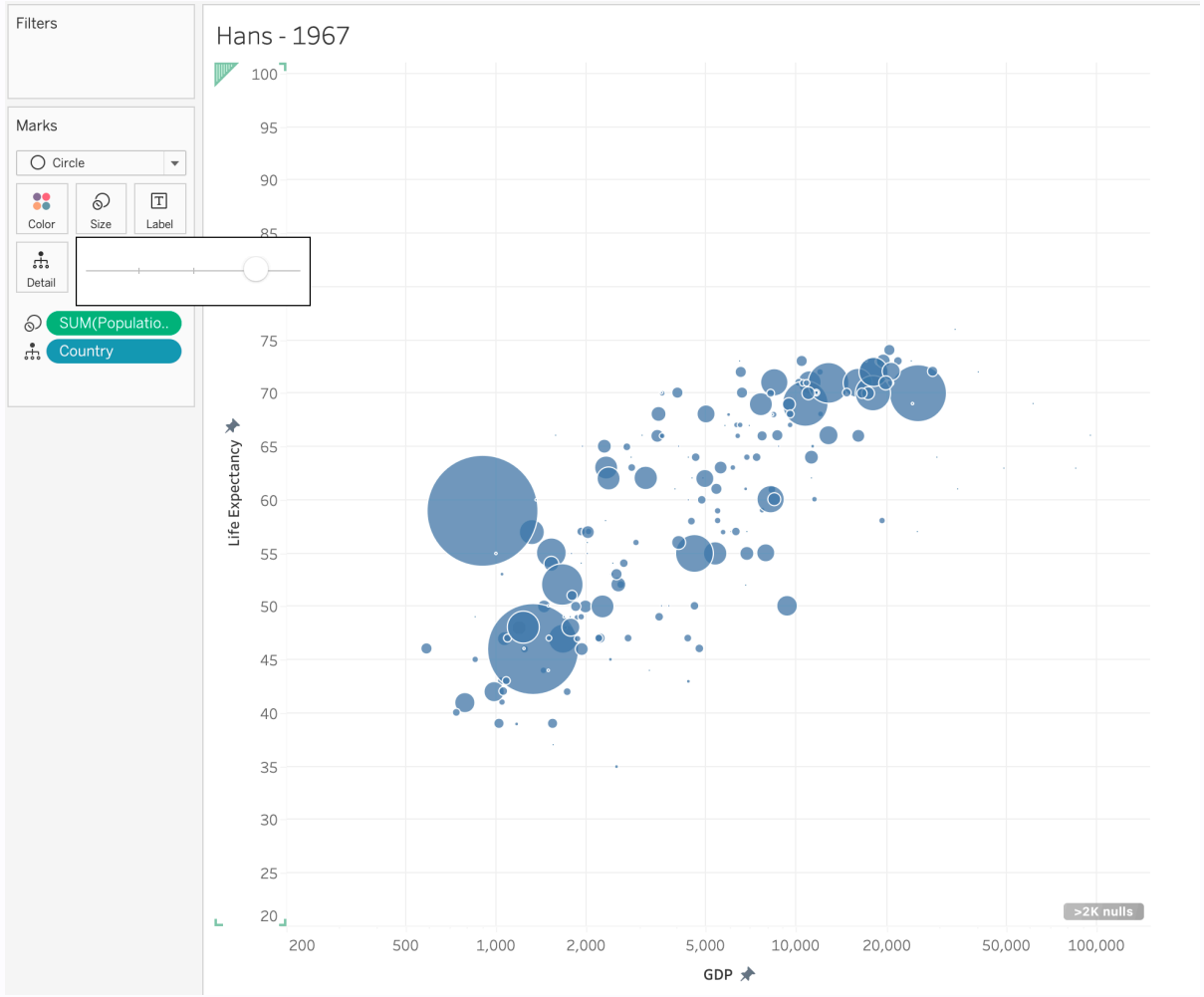
SUM(Populatio..

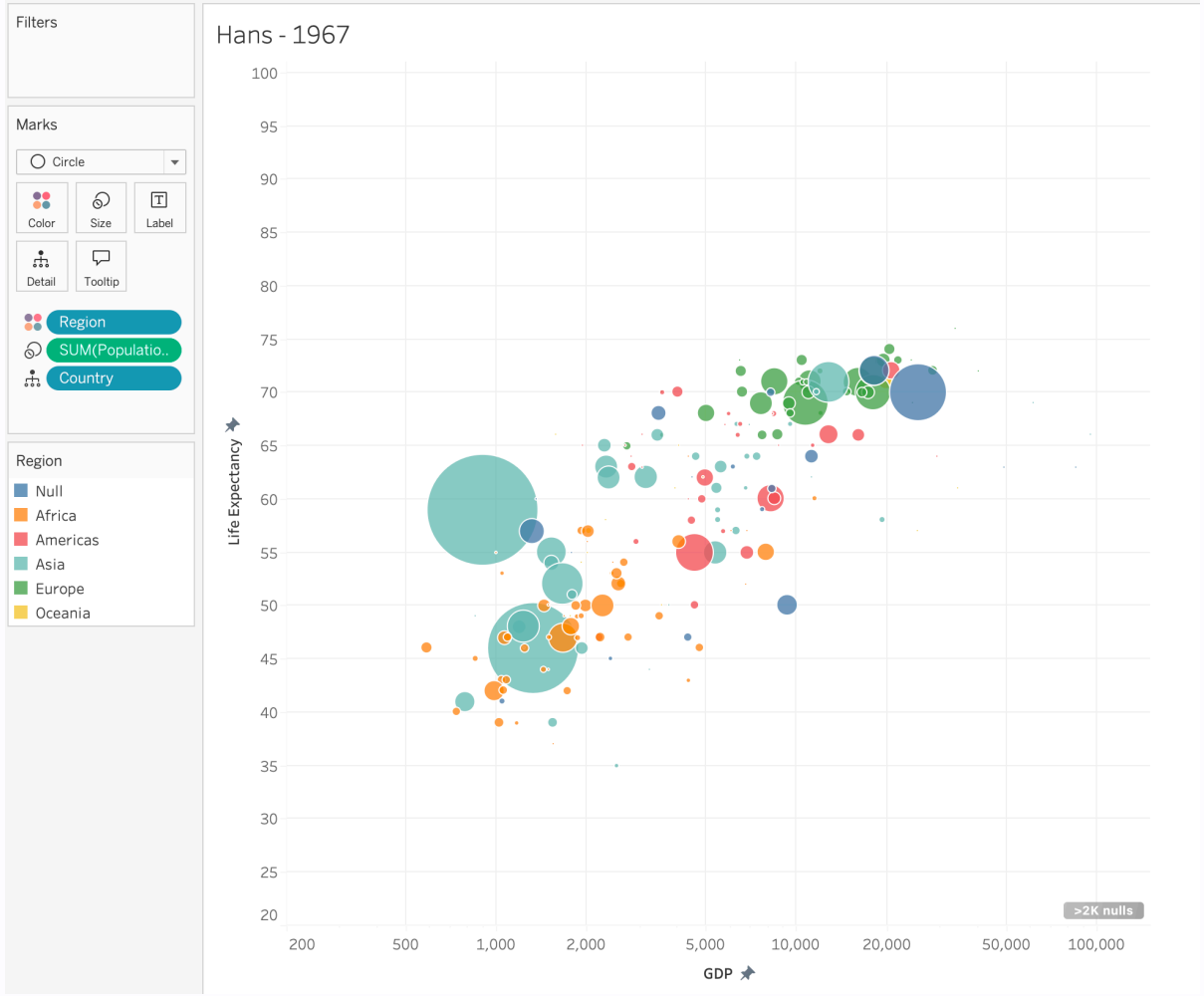
Country

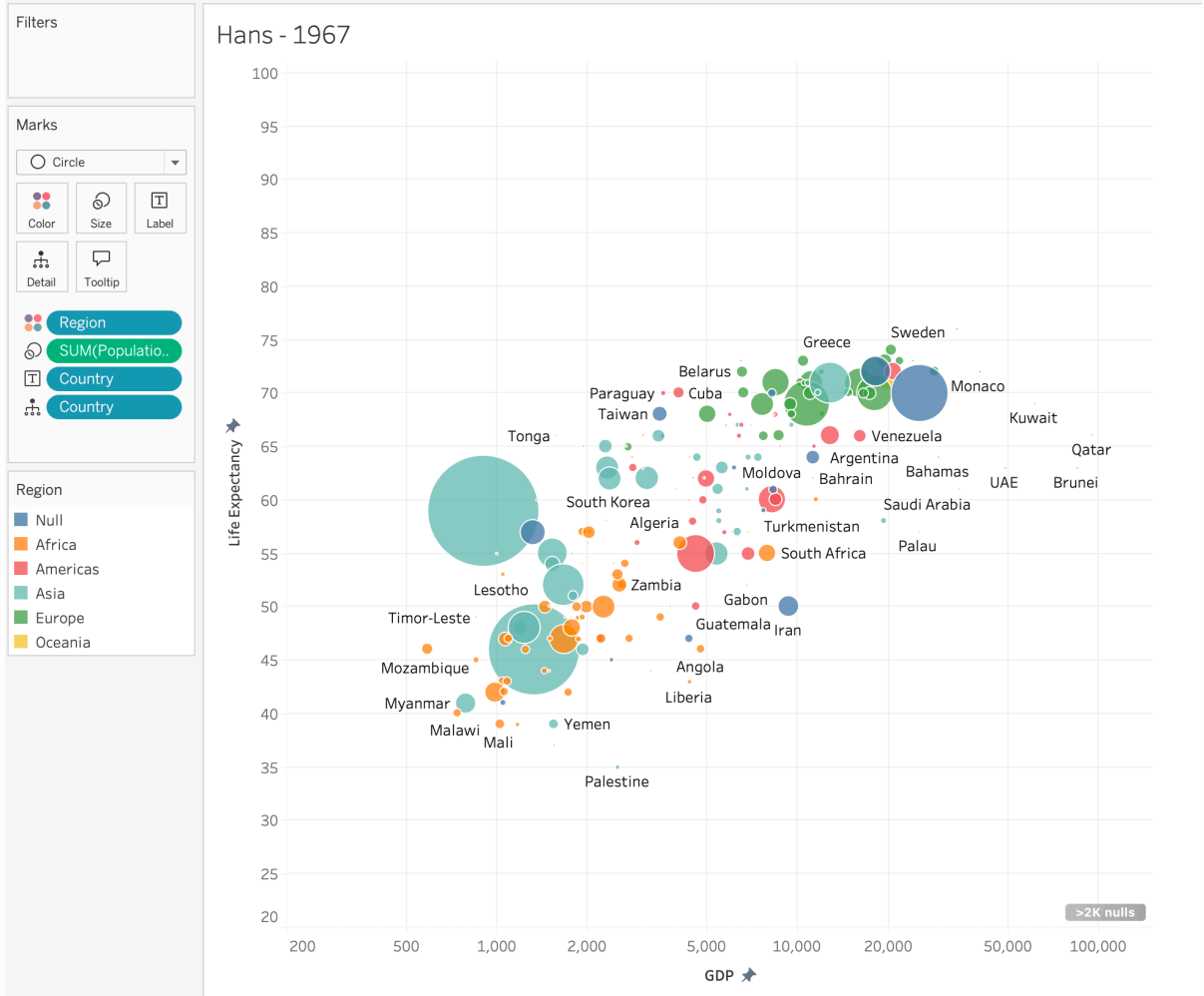
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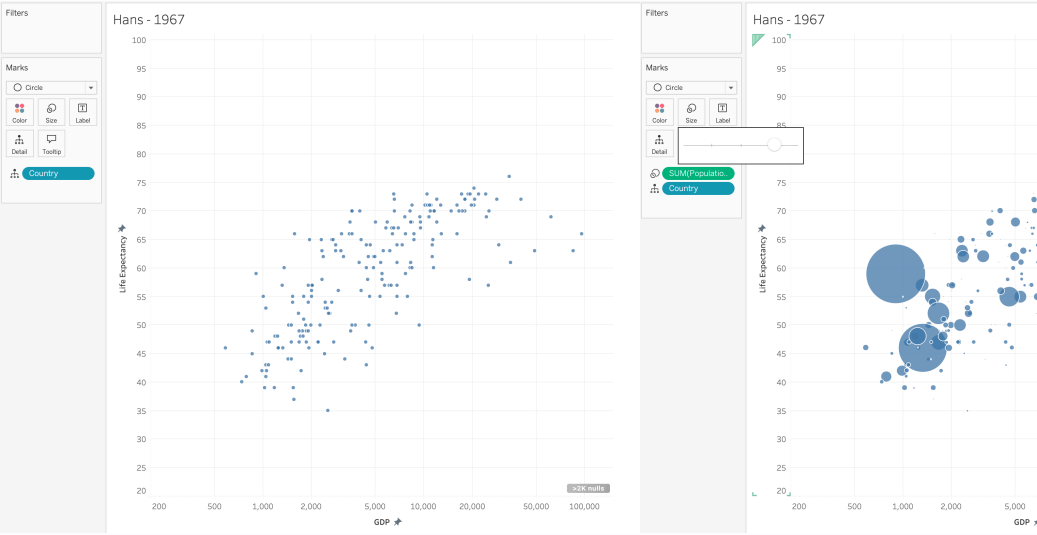


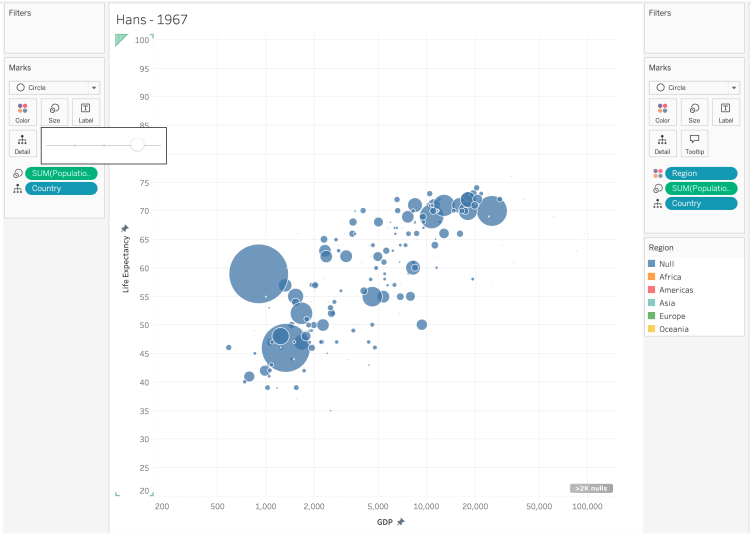

Filters
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Detail
Tooltip
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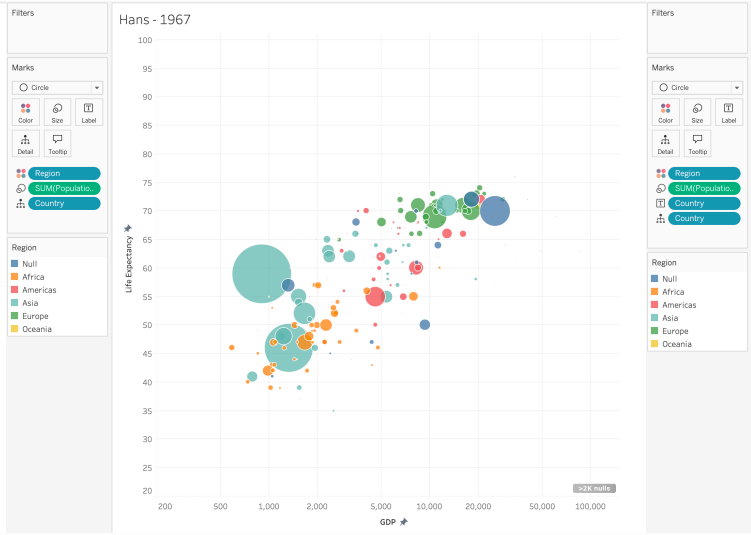

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Life Expectancy
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GDP
>2K nulls


Filters
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Life Expectancy
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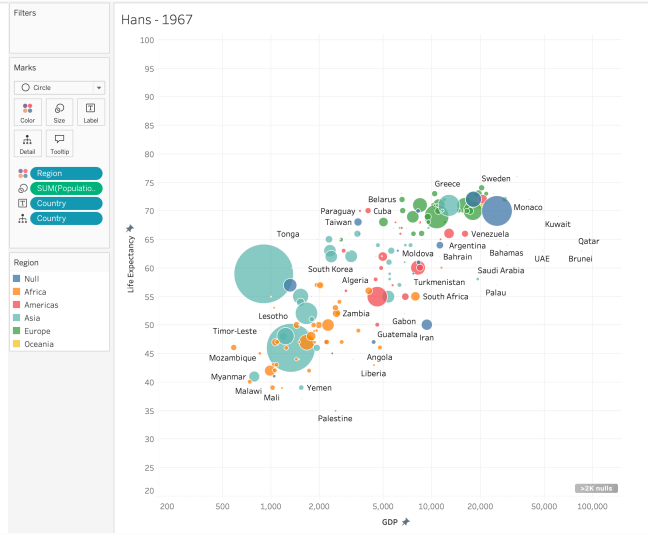

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Europe
Oceania
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Sweden
Belarus
Monaco
Paraguay
Cuba
Taiwan
Kuwait
Tonga
Venezuela
Qatar
Argentina
Moldova
Bahrain
Bahamas
UAE
Brunei
South Korea
Saudi Arabia
Algeria
Turkmenistan
Palau
South Africa
Zambia
Lesotho
Gabon
Timor-Leste
Guatemala
Iran
Mozambique
Angola
Liberia
Myanmar
Malawi
Yemen
Mali
Palestine
>2K nulls


Filters
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Hans - 1967
Life Expectancy
GDP
>2K nulls


Filters
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Country
Hans - 1967
Life Expectancy
GDP
>2K nulls


Filters
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Asia
Europe
Oceania
Hans - 1967
Life Expectancy
GDP
>2K nulls

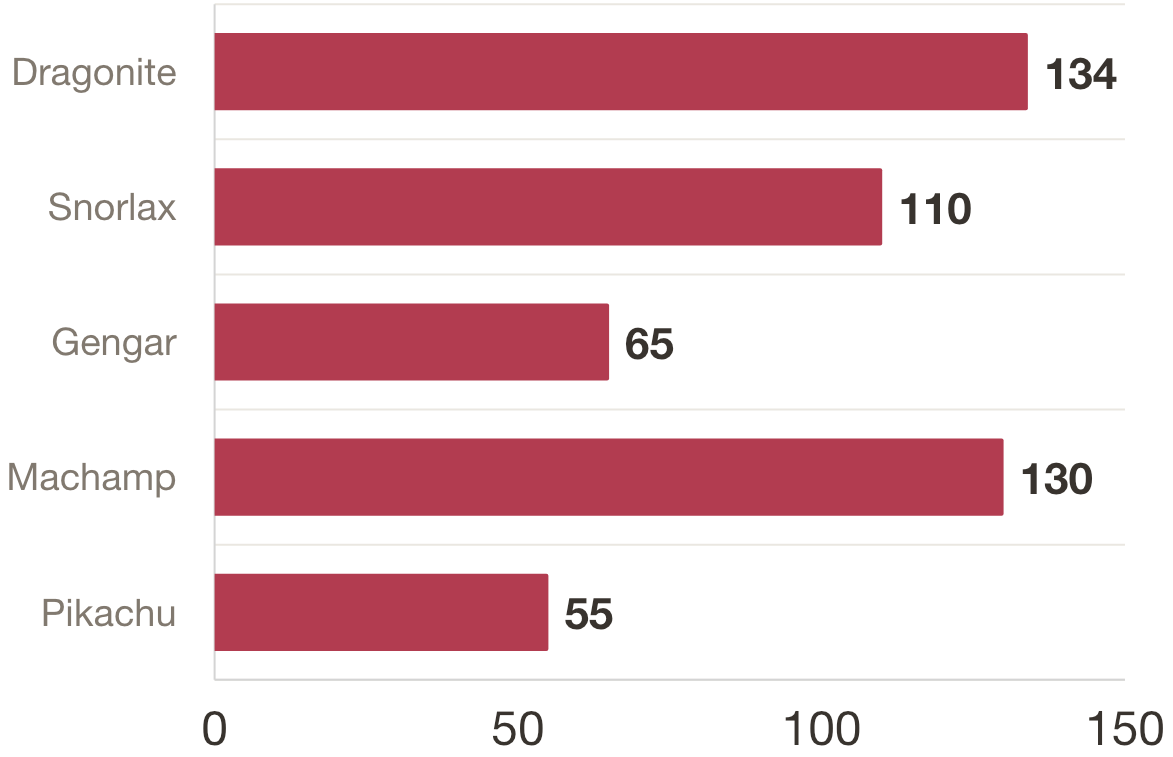
Filters
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Hans - 1967
Life Expectancy
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Hans - 1967
Life Expectancy
GDP
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Greece
Sweden
Belarus
Monaco
Paraguay
Cuba
Taiwan
Kuwait
Tonga
Venezuela
Argentina
Qatar
Moldova
Bahrain
Bahamas
UAE
Brunei
South Korea
Saudi Arabia
Algeria
Turkmenistan
Palau
South Africa
Lesotho
Zambia
Gabon
Timor-Leste
Guatemala
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Liberia
Myanmar
Malawi
Yemen
Mali
Palestine

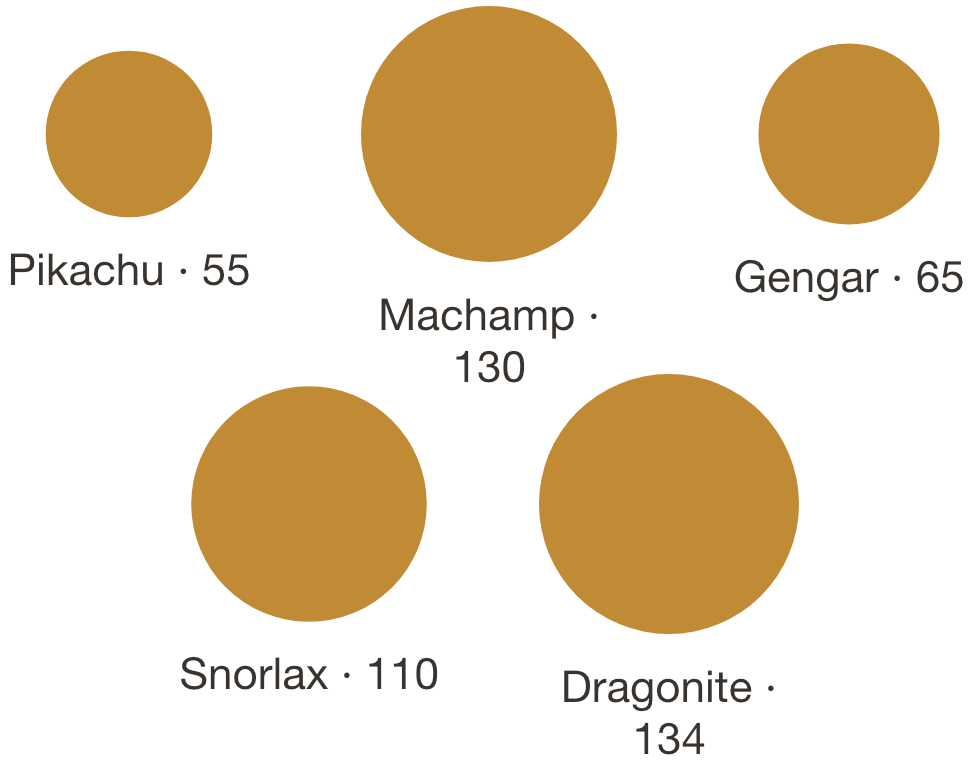
Which comparison is easier?

Same Pokémon. Same Attack values.

POSITION + LENGTH



AREA

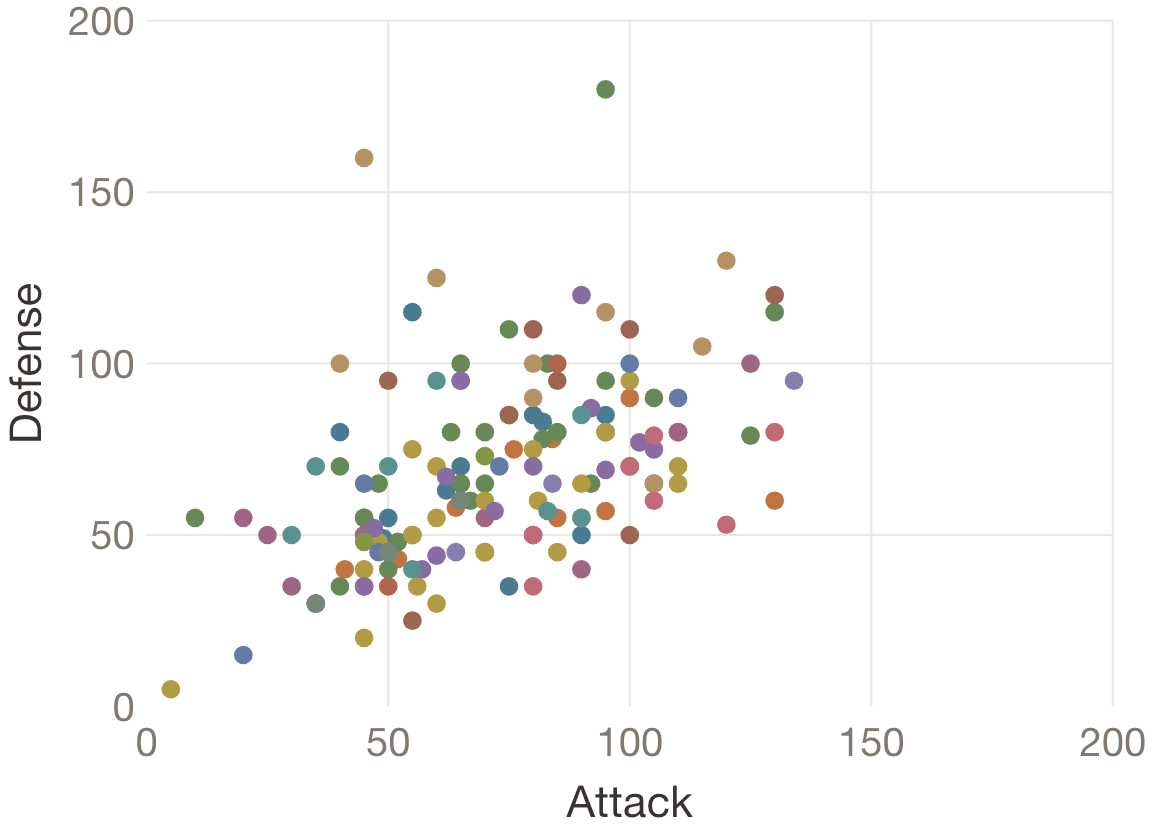


5 selected species · base Attack

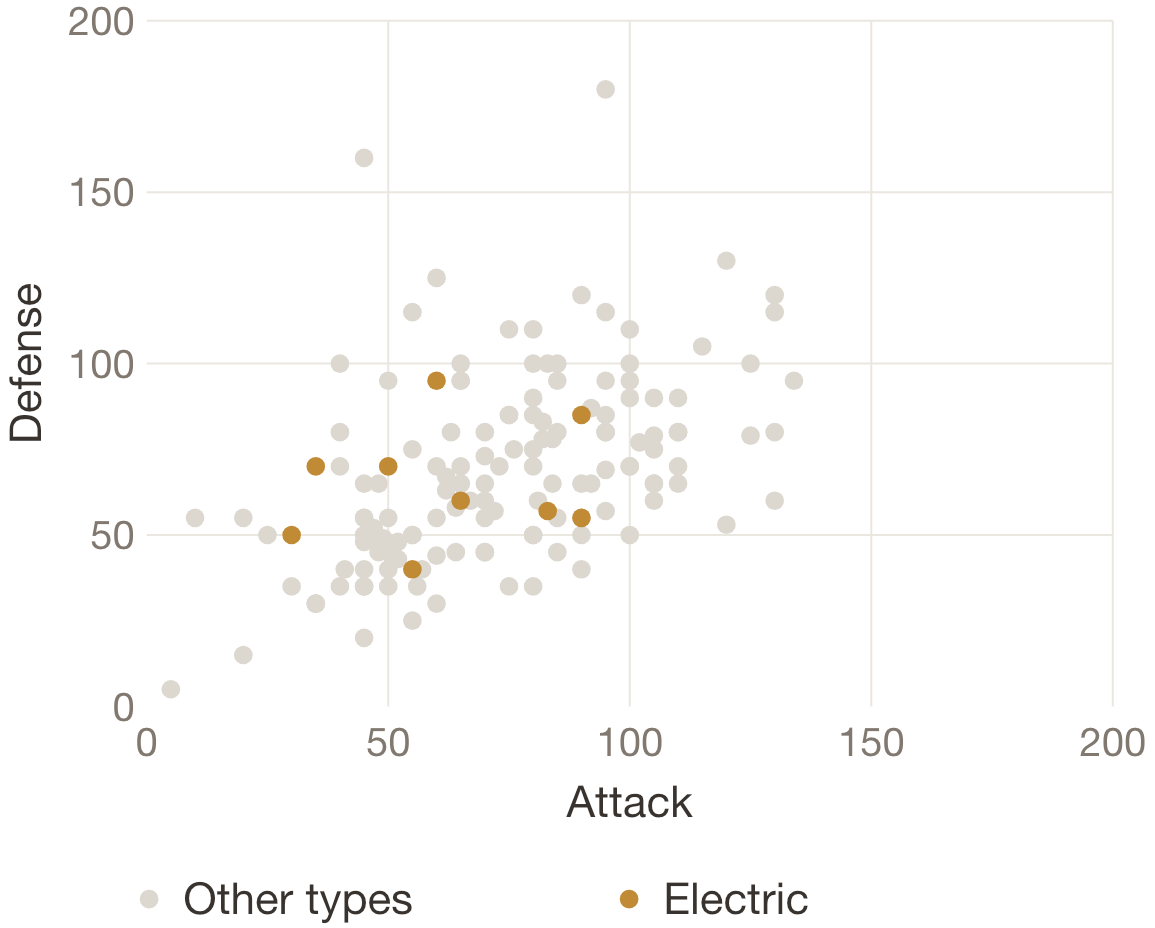
What catches your eye first?

Color everything — or direct attention?

ALL TYPES



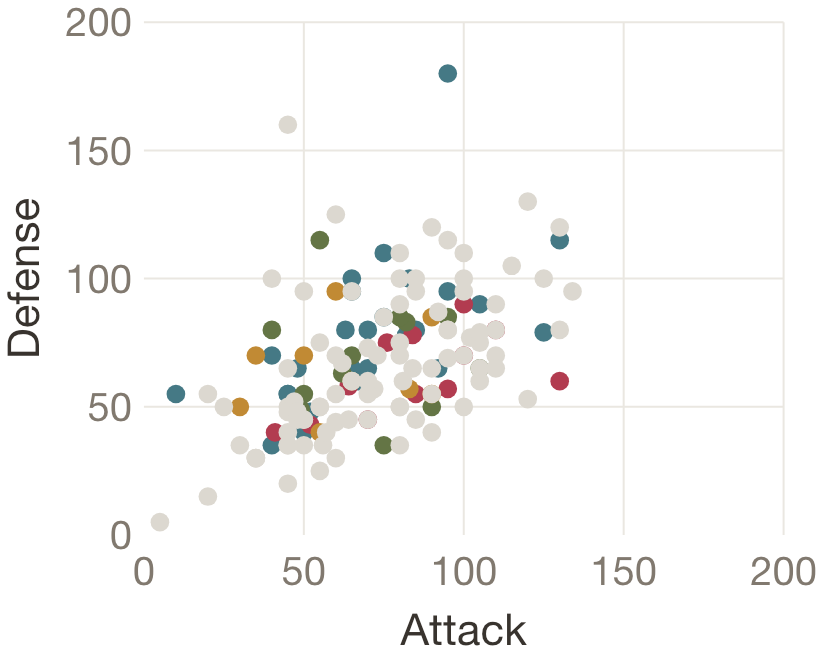
ELECTRIC IN FOCUS



Color needs a job.

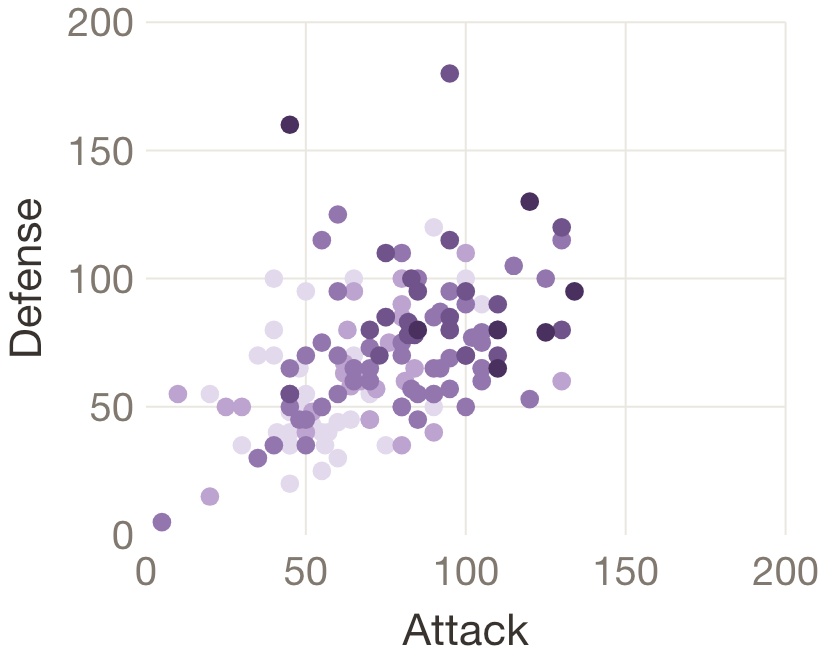
Different kinds of data need different color logic.

TYPE



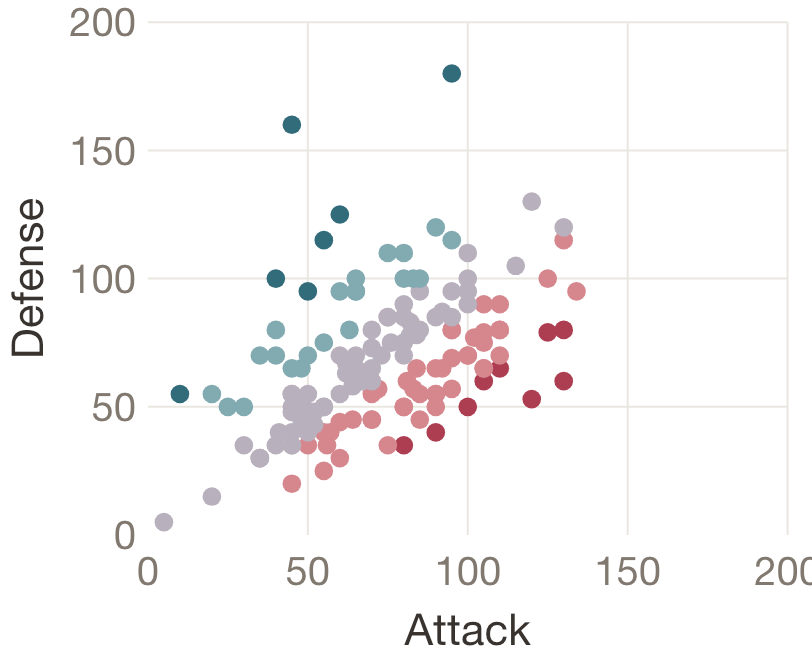
- Water
- Fire
- Grass
- Electric
- Other

WEIGHT



- 0–<10 kg
- 10–<30 kg
- 30–<70 kg
- 70–<150 kg
- 150+ kg

DEFENSE – ATTACK



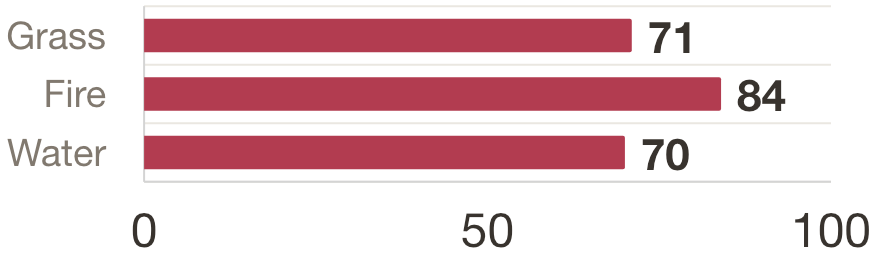
- ≤–41
- –40…–11
- –10…10
- 11…40
- ≥41

Layout creates relationships.

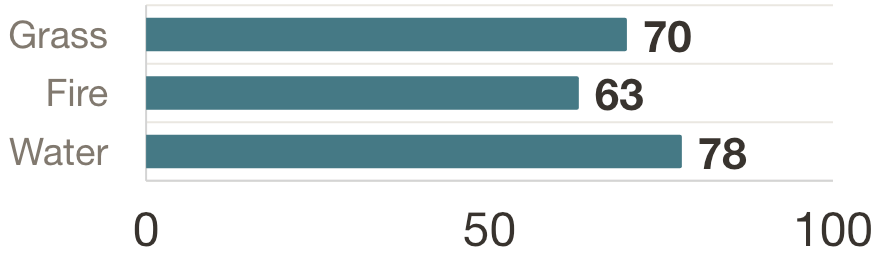
Same values. Different alignment.

SCATTERED

Attack

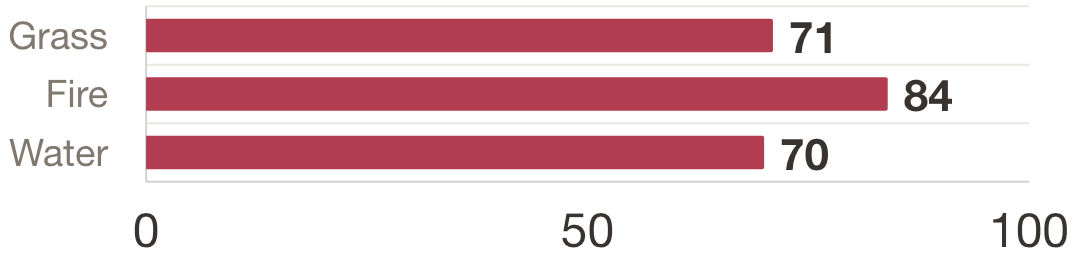


Defense

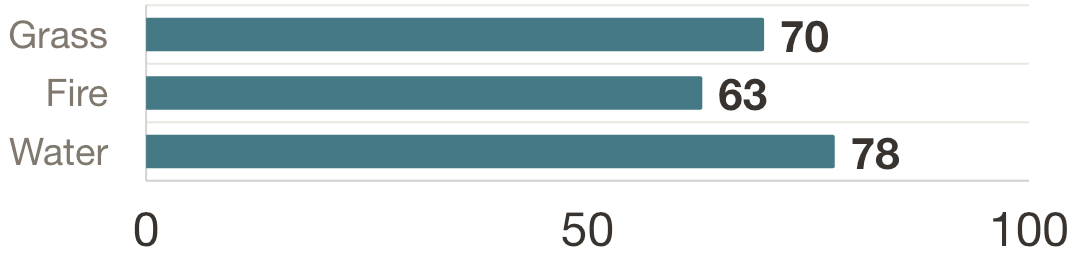


ALIGNED

Attack



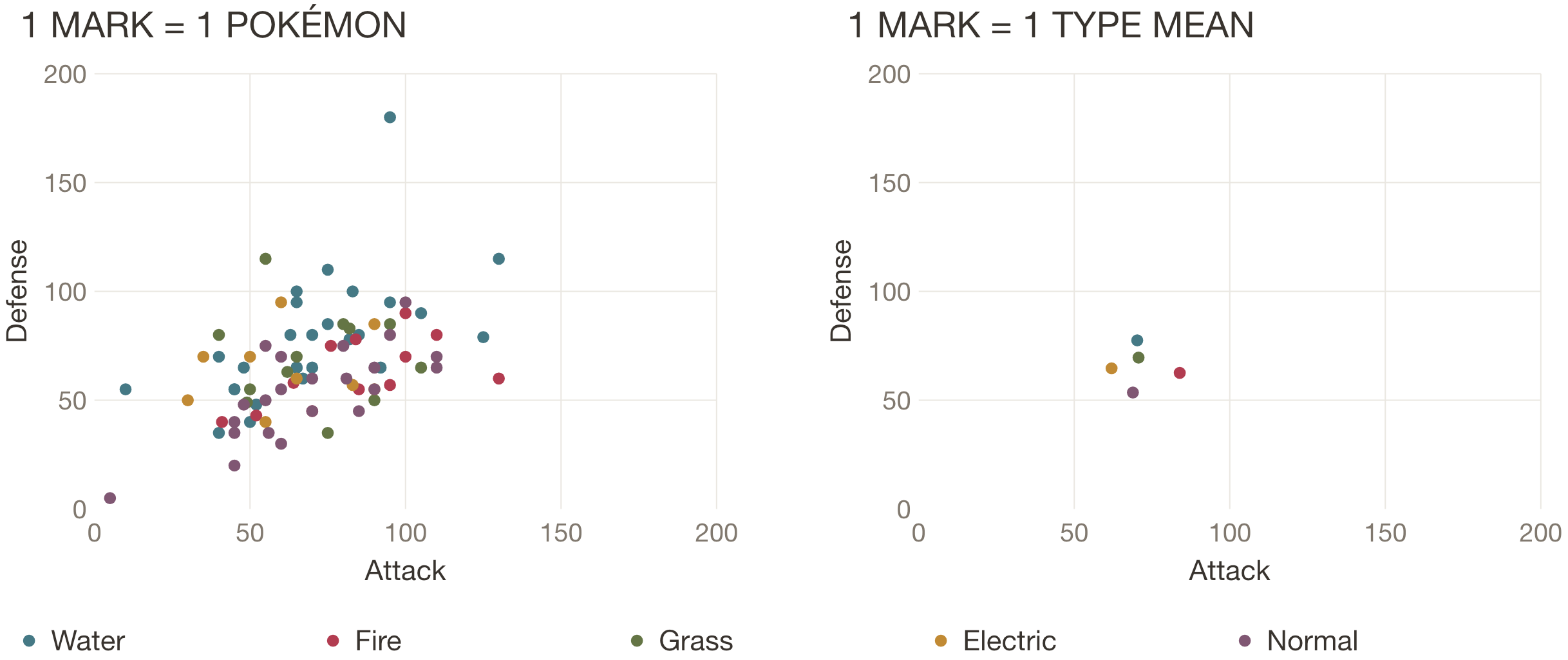
Defense



Mean base stats · primary type · rounded to whole numbers

What does one mark represent?

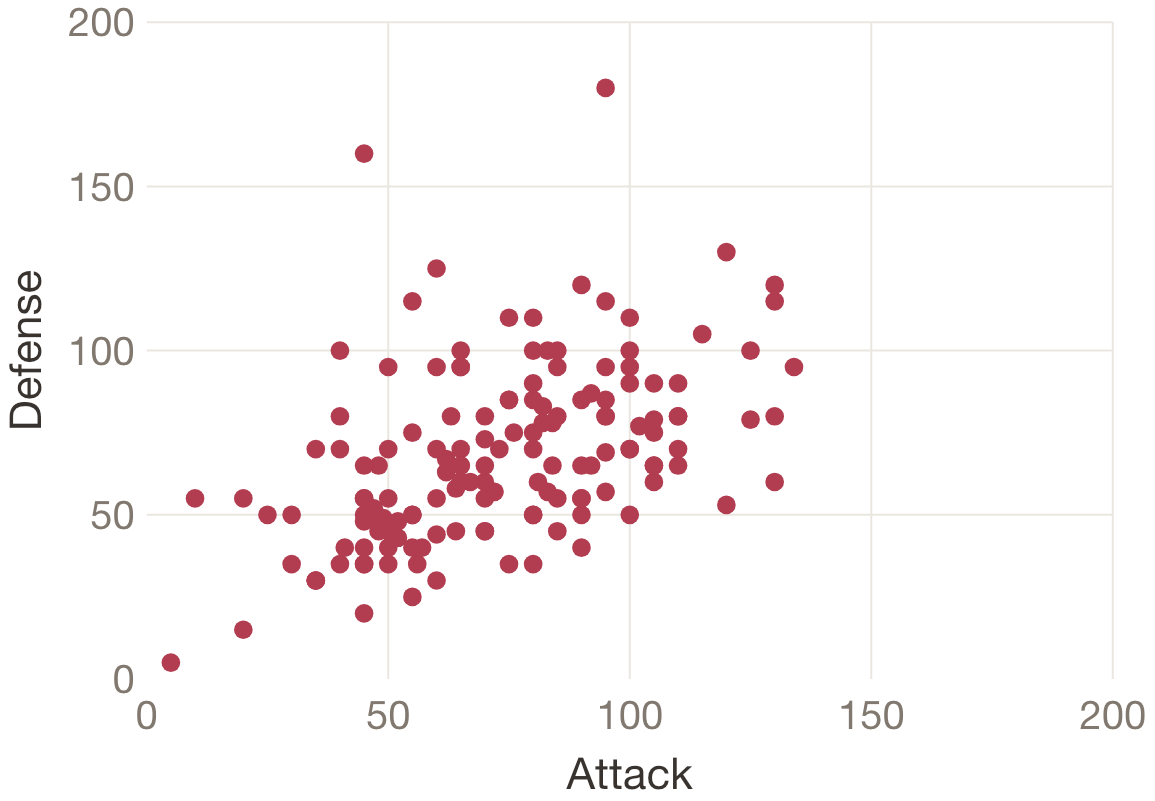
Aggregation changes the story.



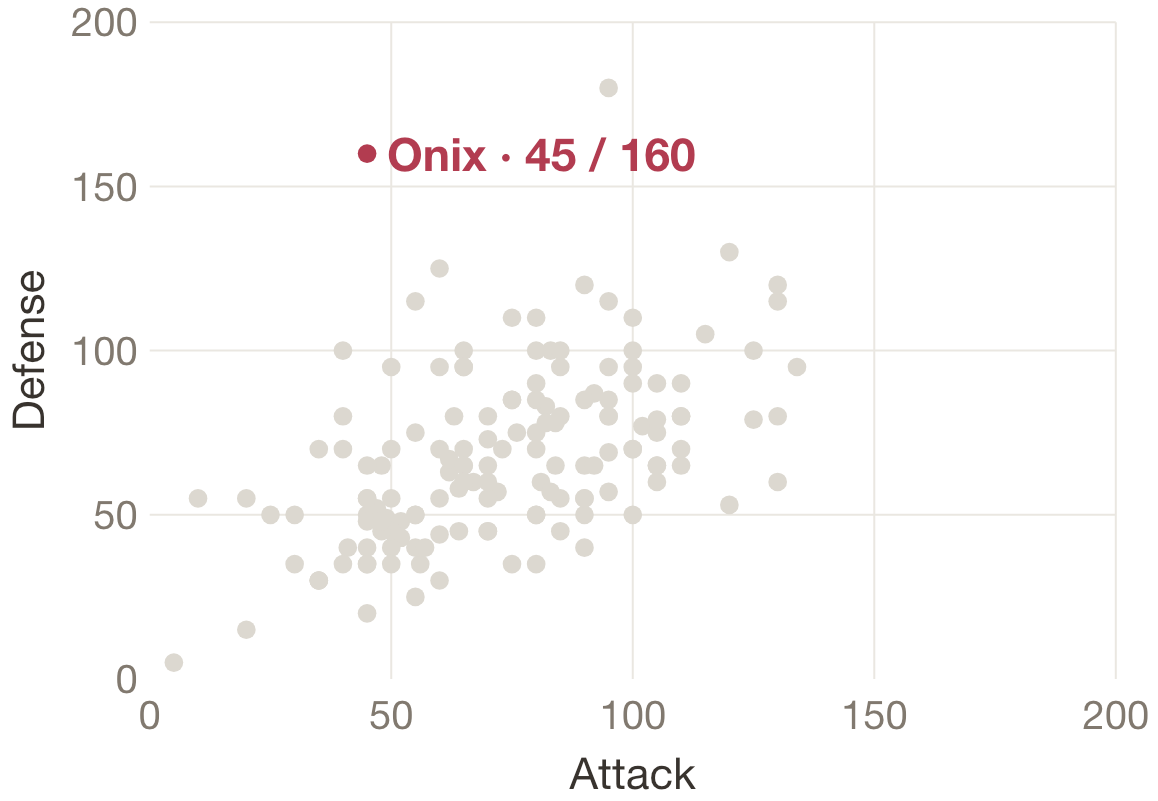
Help the reader find the point.

A title is a chance to say something.

Pokémon stats



Onix: high Defense, low Attack



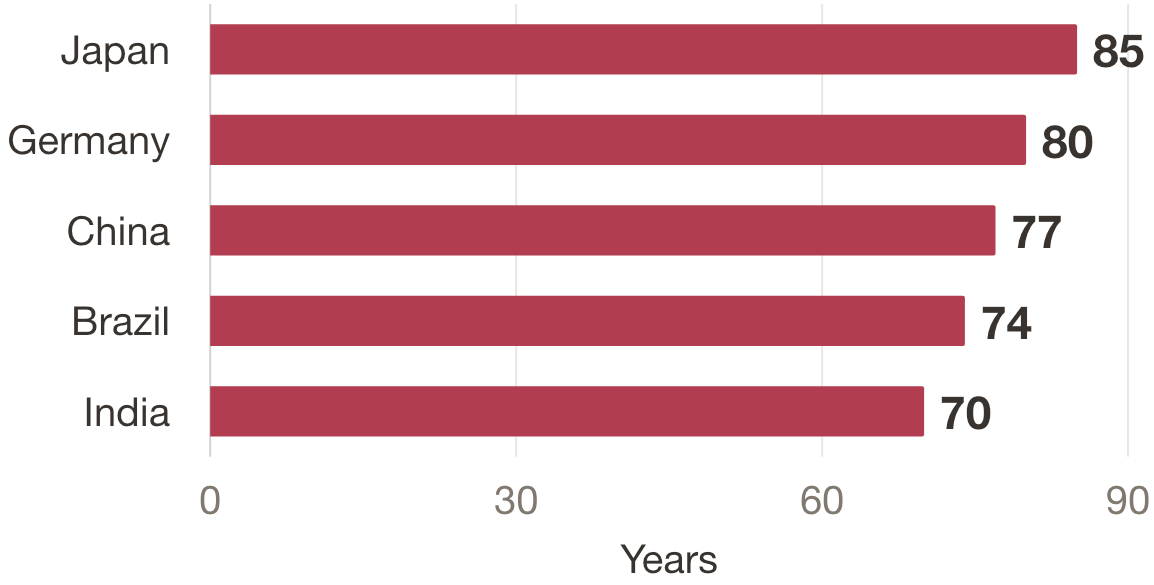
ONE CLEAR CLAIM. ONE USEFUL LABEL.

Tableau chart types: comparisons

Life expectancy in five countries (2020)

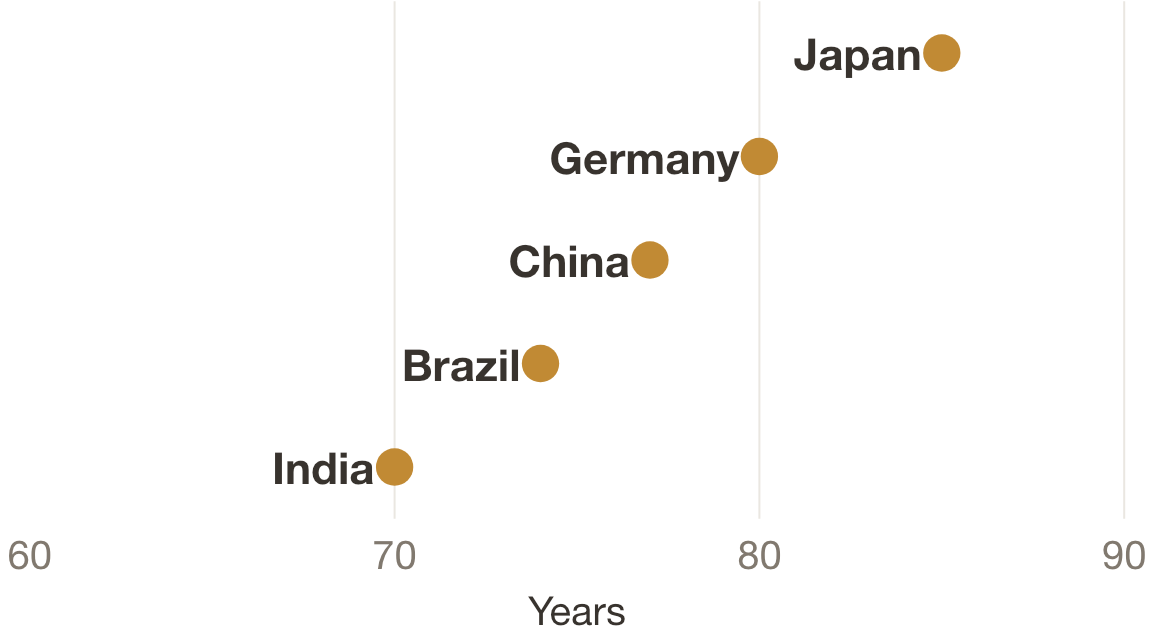
Bar chart

Lengths compare values. Start the scale at zero.



Dot plot

Positions make small differences easier to see.



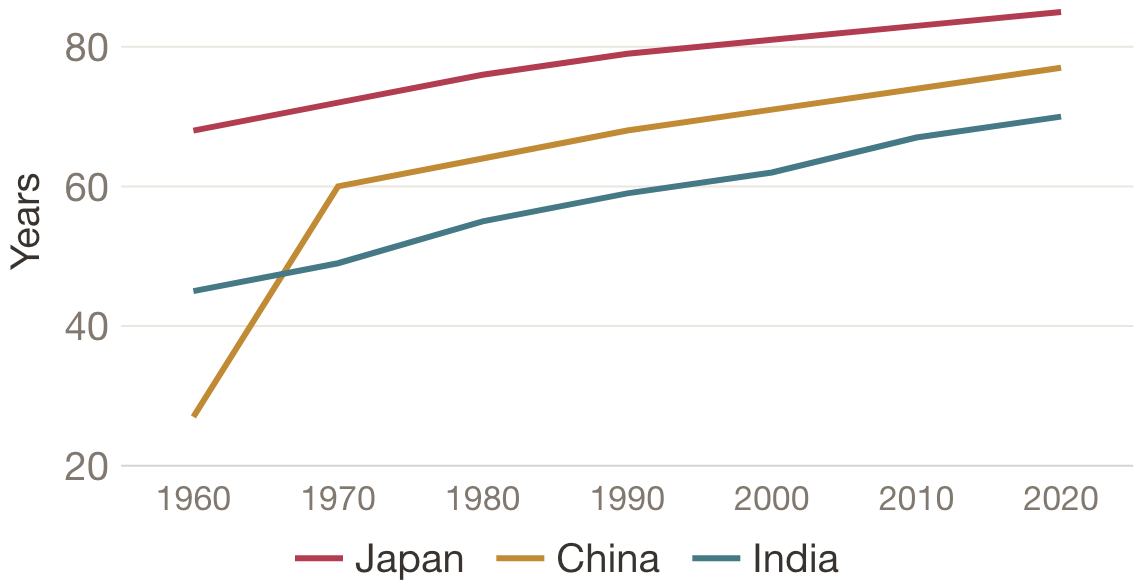
Select fields, then open Show Me. The options depend on your fields.

Tableau chart types: change over time

A line shows a trajectory. An area also emphasizes magnitude.

Line chart

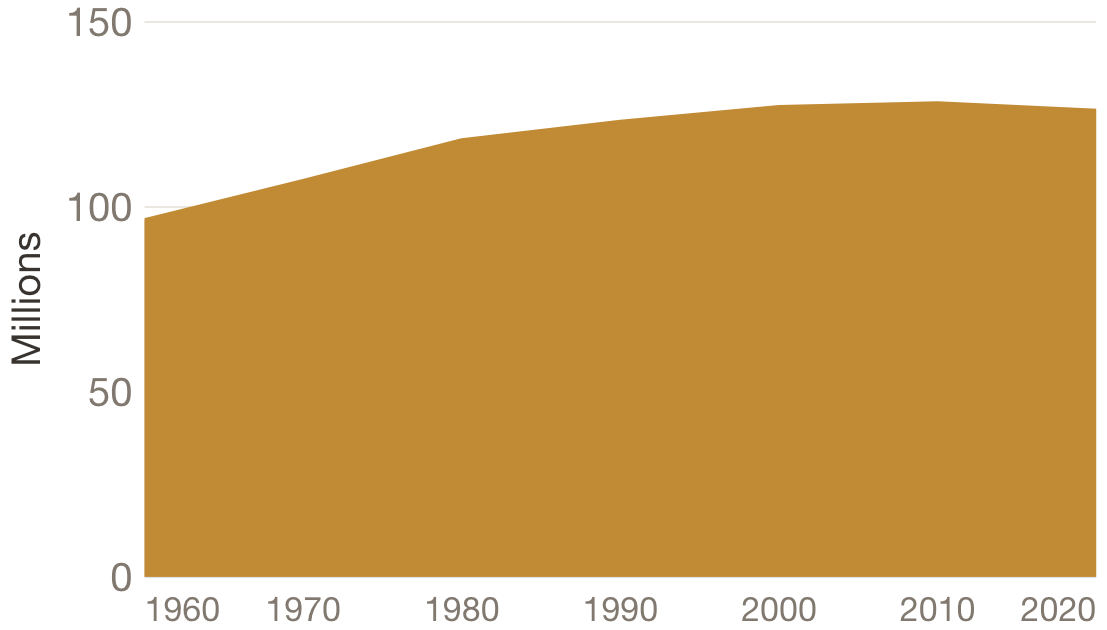
Life expectancy in Japan, China and India



Line: compare trends over time.

Area chart

Japan's population, in millions



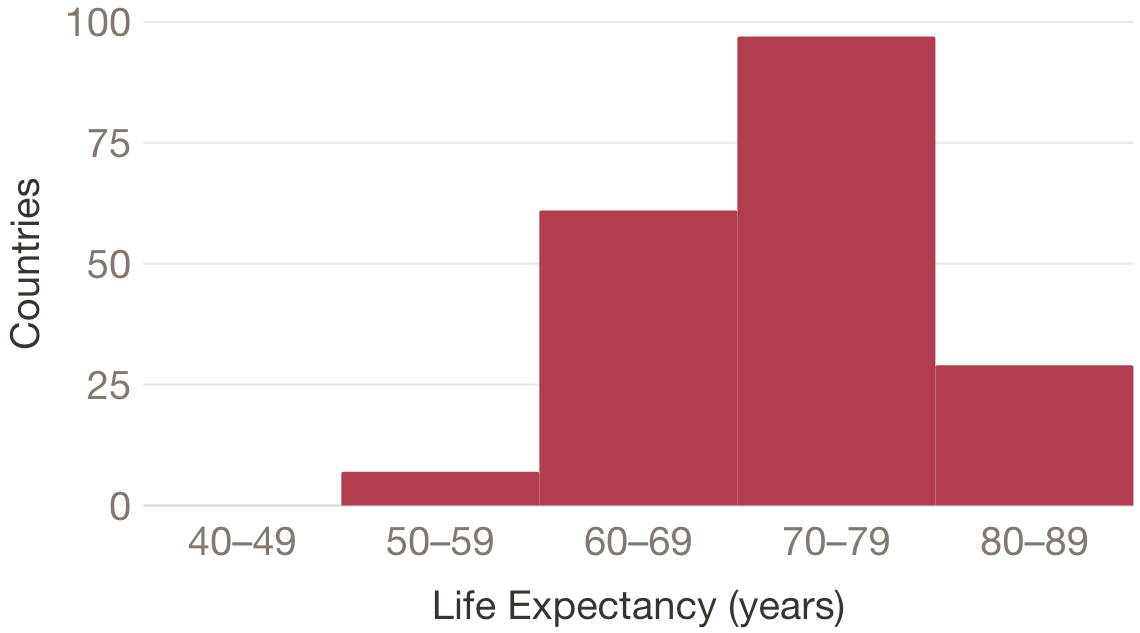
Area: use a meaningful zero baseline.

Tableau chart types: distributions

What is typical, and how much variation is there?

Histogram

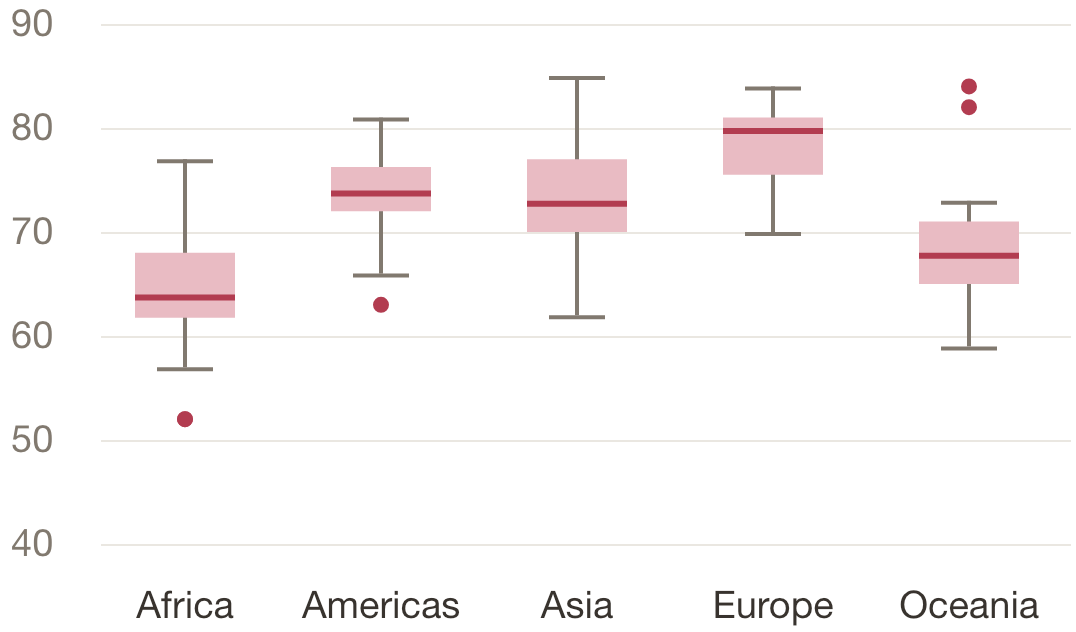
How many countries fall in each range? (2020)



A histogram counts values in numeric bins.

Box-and-whisker plot

Country life expectancy within each region (2020)



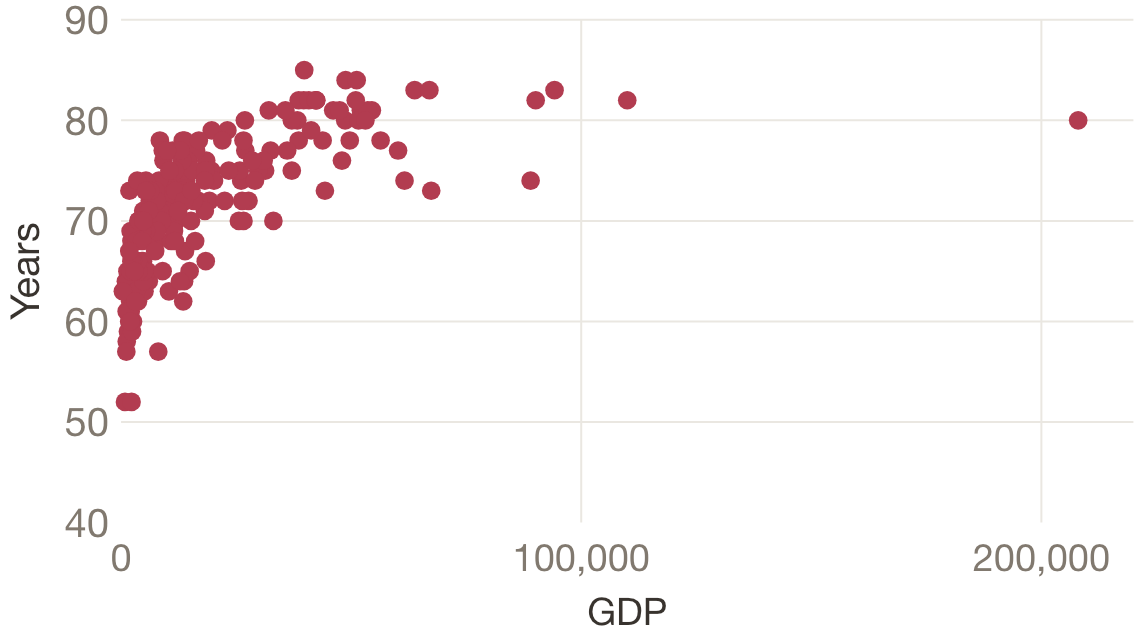
Median, middle 50%, spread and outliers.

Tableau chart types: relationships and patterns

One pair of measures, or a matrix of repeated comparisons

Scatter plot

GDP and life expectancy, one circle per country



Association does not establish cause.

Highlight table

Life expectancy by country and decade

	1980	1990	2000	2010	2020
Japan	76	79	81	83	85
Germany	73	75	78	80	80
China	64	68	71	74	77
Brazil	62	67	71	73	74
India	55	59	62	67	70

40 50 60 70 80 90

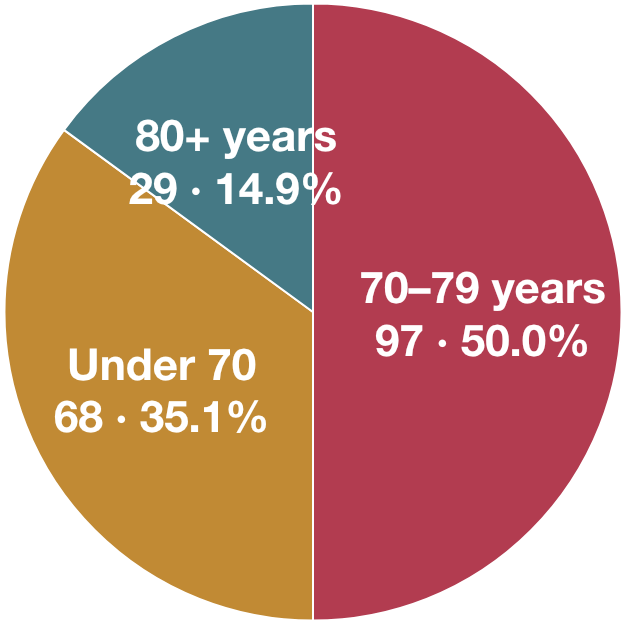
One shared color scale, in years.

Tableau chart types: parts of a whole

How many countries fall in each life-expectancy band? (2020)

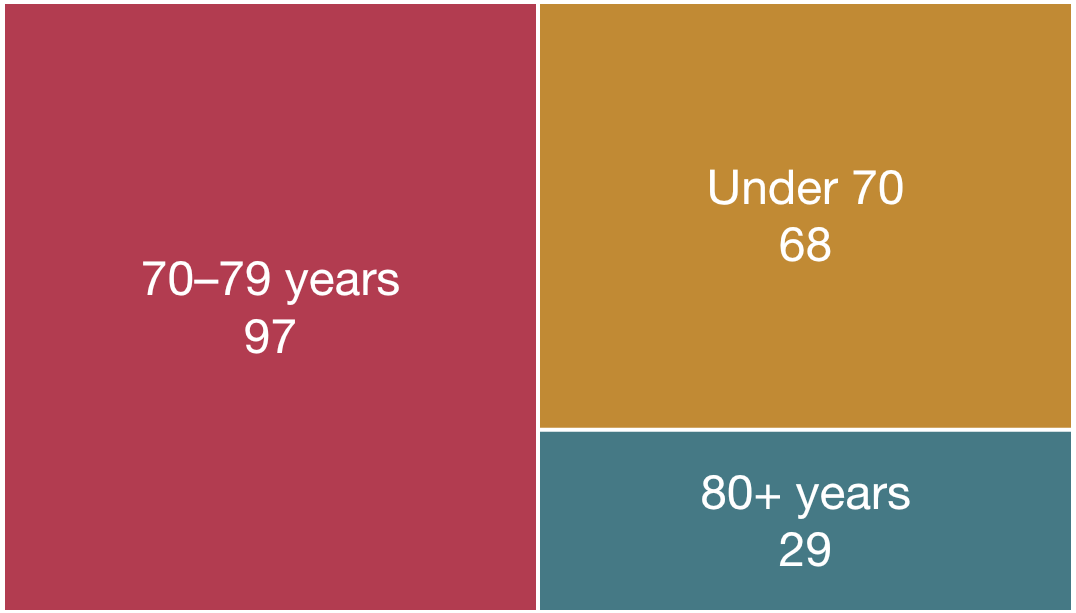
Pie chart

Shares of a total



Treemap

Parts sized by area



194 countries with valid life expectancy. Shares of countries, not people.

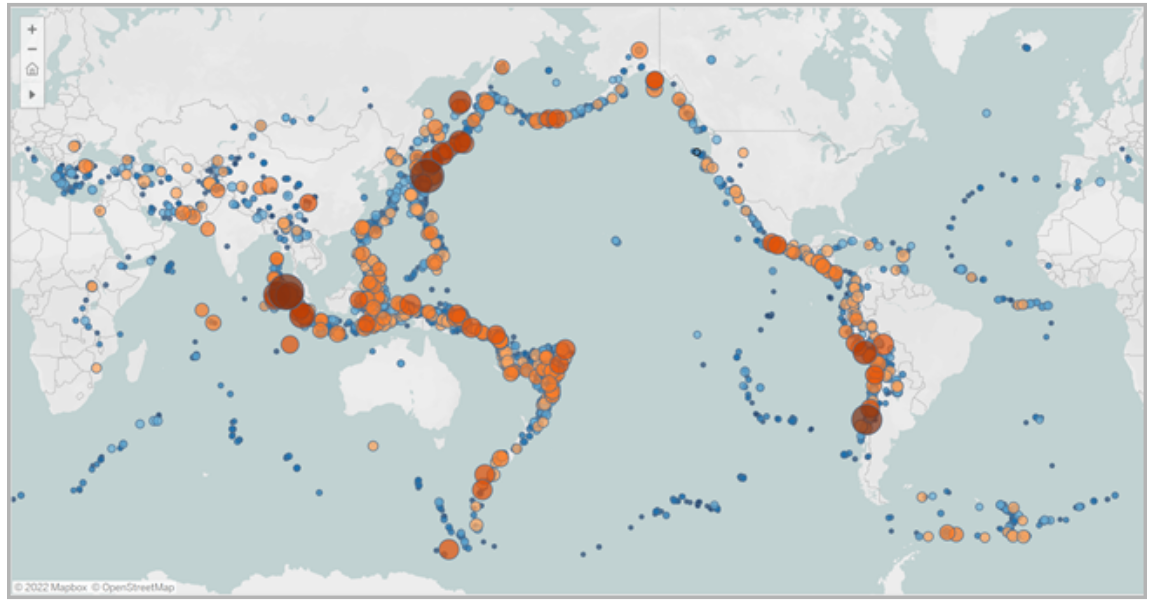
Life band → Color COUNTD(Country) → Angle / Size

Tableau chart types: geography

A map answers a question about location

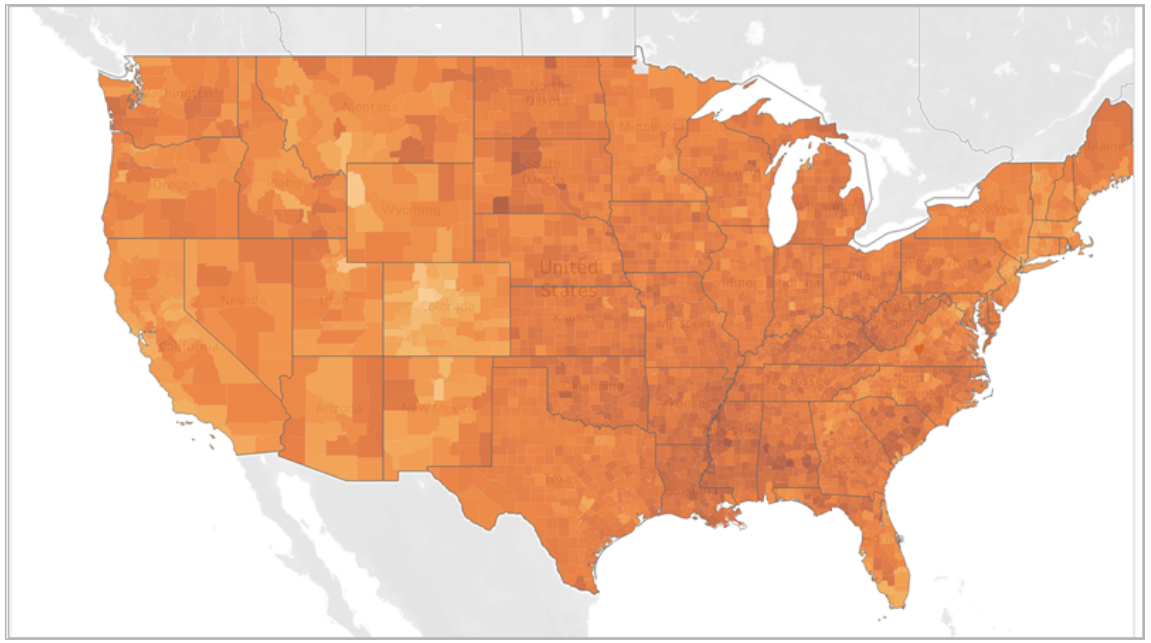
Symbol map

Size represents a value at each location.



Filled map

Color represents a value for each area.

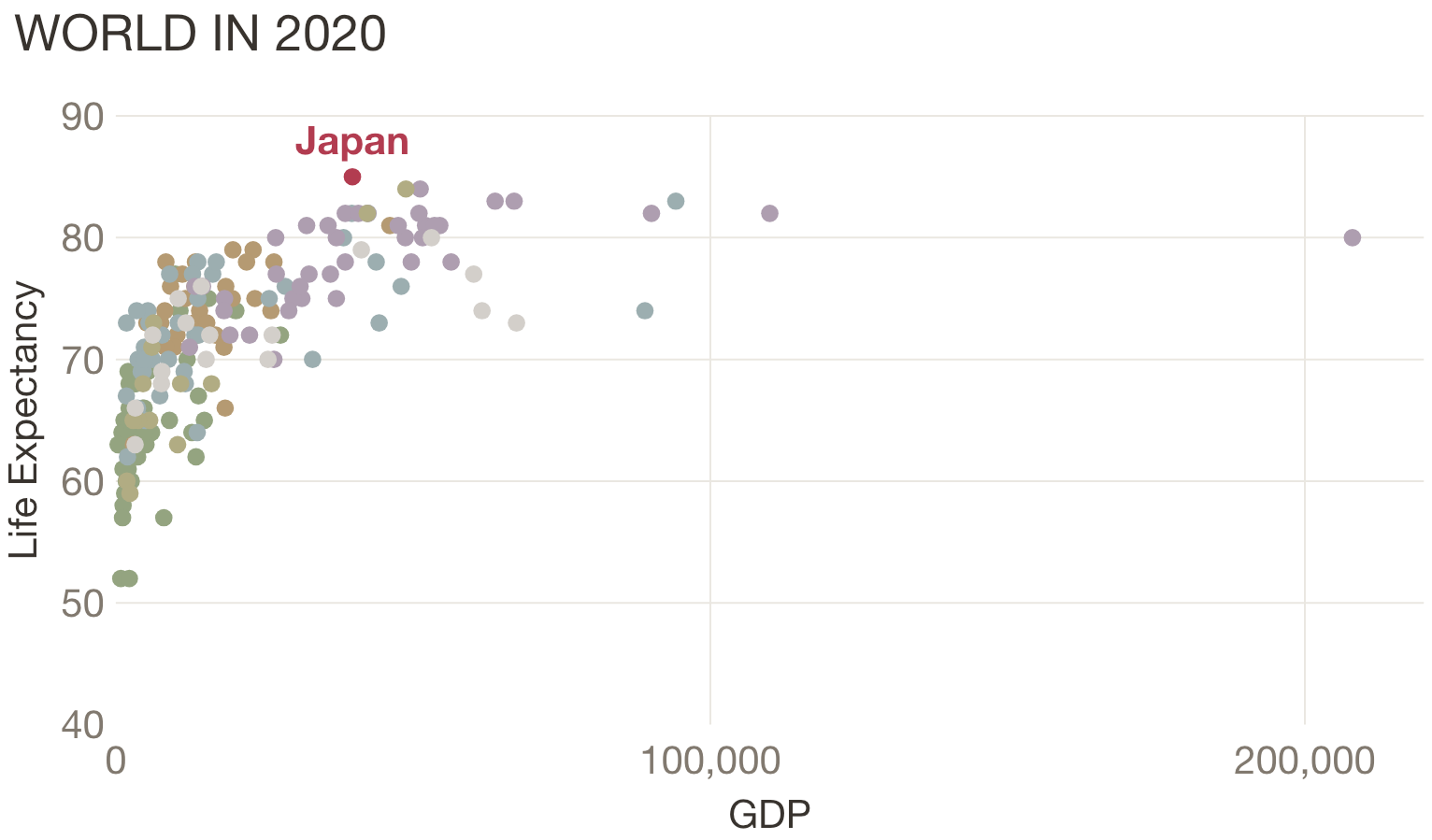


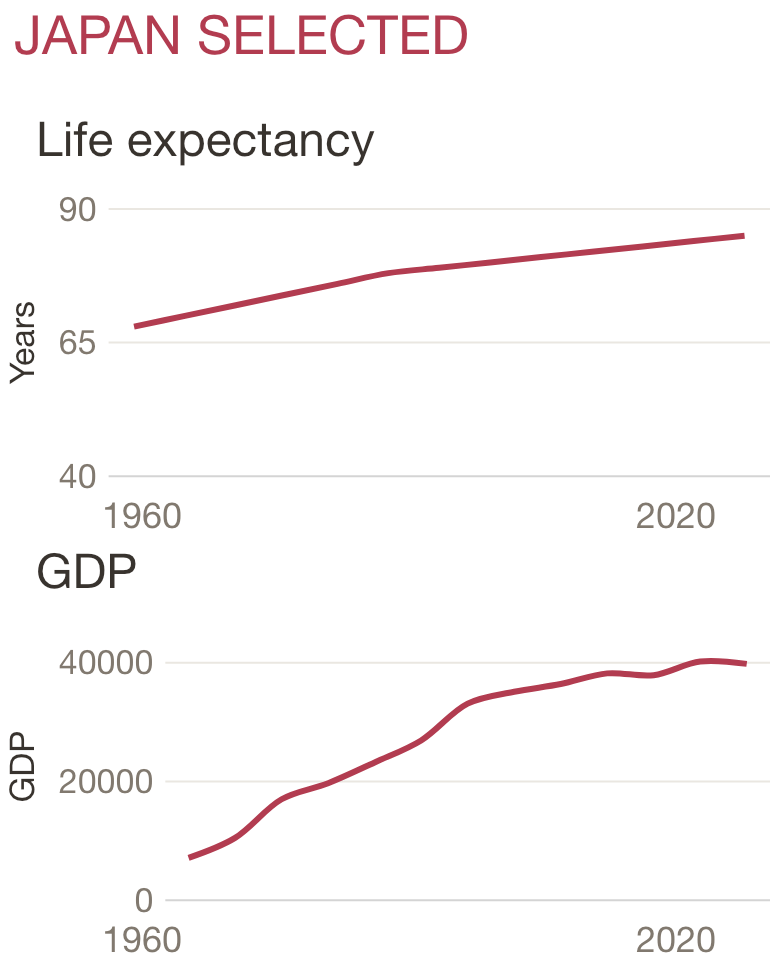
Try hans.csv: Country as a geographic field; Year = 2020; Life Expectancy on Color.

Map images: Tableau documentation examples

Today's tip: a Tableau dashboard

Last week's Gapminder chart, connected to a country's history



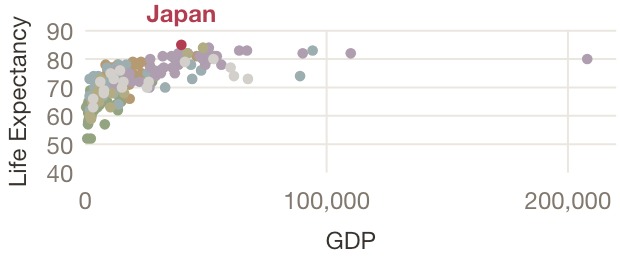


Click a country in the overview. Explore its history on the right.

The three worksheets

Use the same hans.csv connection for every sheet

1 World

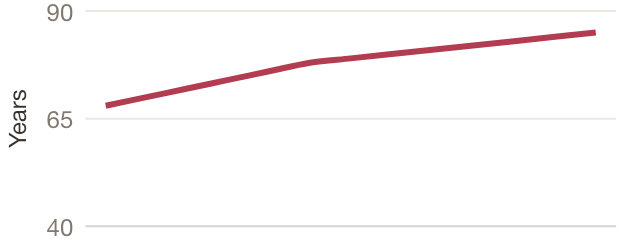


Columns AVG(GDP) Rows AVG(Life Expectancy)

Circle Country: Detail region: Color Year: 2020

Year goes in Filters, not Pages. Keep one circle per country.

2 Life History

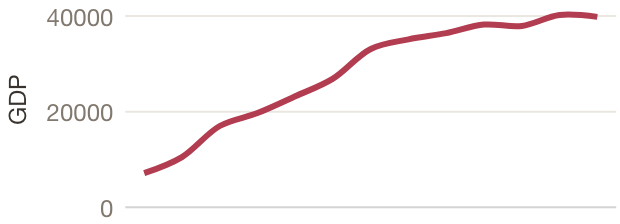


Columns Year (continuous) Rows AVG(Life Expectancy)

Line Country: Color Year: 1960–2020

Convert Year to Dimension, then Continuous. Keep 1960–2020.

3 GDP History



Duplicate Life History.
Replace the measure with AVG(GDP).

Same countries. Same years. A different measure.

The dashboard layout

A large overview and a smaller history column

1200 × 800



Tiled: horizontal container, with a vertical container on the right

1 New Dashboard

Set Size to Fixed: 1200 × 800.

2 Tiled layout

Drag World left. Stack both histories on the right.

3 One year control

Year filter menu:
Apply to Worksheets
Selected Worksheets: World only.

Keep both histories at 1960–2020.

A country click keeps the whole history

Dashboard > Actions > Add Action > Filter

Source

World

Run on

Select

Targets

Life History + GDP History

Selected Fields

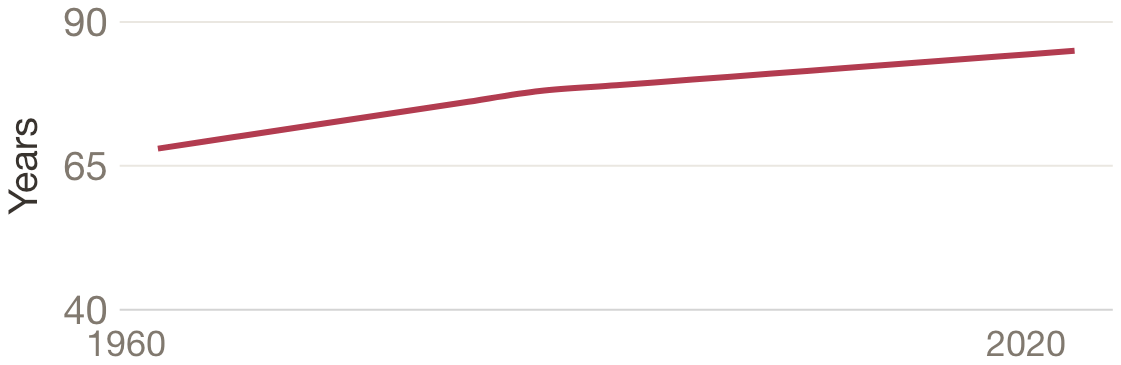
Country → Country

Clear selection

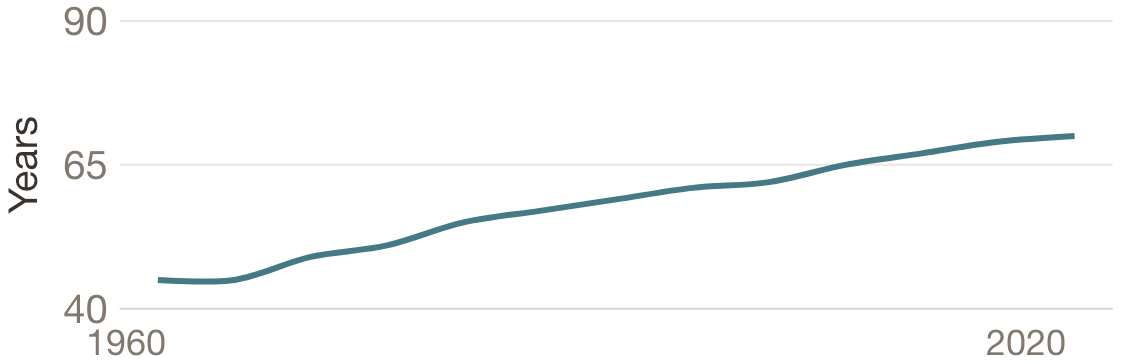
Show all values

Do not pass Year to the history sheets.

Click Japan



Then click India



Life expectancy, 1960–2020